

Draft Oak Savanna Monitoring Standard Operating Procedures: For Field Testers



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DRAFT

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Oak Savanna Monitoring Standard Operating Procedure (SOP)

SOP 1: Species Richness Meandering Walk

The meandering walk is used to track the progress of a monitoring area by attempting to record all species expressed and general conditions in the management unit in a given year and through time. The observer strives to develop as complete a plant species list (herbaceous and woody) as possible for a monitoring area during each meandering walk. This list will be larger than would be possible using a plot- or transect-based method. Expression of exotic and/or invasive species can provide insights about threats through time. In addition, canopy coverage or light levels provide insights about the capacity to support understory savanna flora and changes through time.

Attributes measured

- Species encountered (presence)
- General abundance (sparse, occasional, frequent, very common)
- General distribution (localized, widespread)
- General herbaceous cover (bare, sparse, moderate, dense)
- Woody density (sparse, moderate, dense)
- Duration of the survey (hours, minutes)
- Area surveyed (hectares)

Equipment and supplies

- Map showing management area boundary and soil moisture type boundaries if applicable; satellite imagery or topography is helpful
- GPS unit with map, if used
- Survey protocol
- Datasheets or mobile device equipped for data entry
- Clipboard
- Pencils/pens
- Watch/clock
- List of species observed in prior surveys (if any), if desired
- Field references, such as field guides or cheat sheets (hard copy or digital)
- Hand lens
- Plastic bags/cooler and/or plant press for specimen collection
- Camera to photograph unknown plants and notable characteristics such as tree condition or erosional or natural features.

Survey timing and schedule

Meandering walks are typically done annually and based on phenological cues. In some cases, two walks may be conducted each year to pick up ephemeral and cool-season species early in the year and then warm-season and late blooming species later in the season. Conduct the walk according to the phenological timing and frequency determined for the survey by confirming with the Site Survey Coordinator or by checking archived data for dates and frequency of sampling during previous monitoring visits. Vernal surveys may record greatest species richness, especially when the area is in a highly degraded condition with extensive woody encroachment.

Sources of error

A brief list of sources of error to consider is:

- Inconsistent application of survey methods
- Observer bias in estimations
- Plant misidentification
- Overlooking rare or cryptic plants, especially when not in bloom

Establishment of sampling units

Each discrete monitoring area delineated by soil moisture type (if more than one moisture type exists within the overall management area) is the sampling unit that should be searched. Use the map you created, as well as a GPS unit, to search the area.

Care should be used when monitoring near the boundary of adjacent monitoring areas, because there may be a region of overlap (or a gap in application) of management in a given unit. If monitoring areas are highly interspersed (e.g., with bluff and lowland topography) or if management is applied in very patchy or complicated ways such that monitoring them separately would be impractical, they may be monitored as one area and treated as a single sample unit. Savanna units often include a mosaic of brightly lit areas with few to no trees (eg: fully sunlit areas embedded as an opening within or on the edge of a wooded area), moderately lit areas with dappled sunlight (eg: primarily with open grown oaks), and more densely shaded oak woodlands or other combination of light levels.

Data collection procedures

Within each monitoring area, do a “botanist’s walk” to create a full species list for the area including relative abundance and distribution category for each species. (see Data field details). This entails traversing all noticeable vegetation zones, including parts of the site with distinctively different light levels (from canopy closure, or from openings along trails, roads, utility rights-of-way, areas of different land use history, etc.) moisture types, topography, or other characteristics that could yield unique species. Aerial photos can provide insights about areas with different light levels and potentially different species compositions based on tree density and presence of open grown oaks apparent from the photos. The botanist’s walk and the woody species walk (SOP 2) should generally match. The botanist should be alert to any special characteristics such as concentrations of indicator species characteristic of deep shade (woodland species like columbine (*Aquilegia canadensis*)), dappled sun as filters through oak trees in a spotted or patchy pattern to the understory (savanna species like purple milkweed (*Asclepias purpurescens*)) and full sun (prairie species like lead plant (*Amorpha canescens*)). Note that these species may be marooned in areas that have suboptimal light levels for their growth needs. Record significant species or patches with GPS, on a map or on a data sheet if GPS is unavailable.

Isolated individuals or patches of invasive species of special management concern can be noted with GPS or other mapping techniques.

Botanists are encouraged to follow their instincts with cues from the landscape about where additional species might be found to obtain the most complete species list. Light levels and thus

species expression may change dramatically through time. The biologist or manager may want to guide technicians to be sure to visit certain parts of the monitoring area known to be of interest. Each soil moisture category that occurs in the monitoring area should be searched and **species recorded separately**. During the meandering walk, special features or conditions such as rock outcroppings, streams/rivers, erosion, seeps, slumping soil, trash dumps, utility lines etc. should be noted and mapped with GPS or alternative mapping techniques in the first year observed. Thereafter, watch for changes in condition (such as a deepening erosion incision) if they occur since the last meandering walk. Photographs will be helpful if different people conduct the walk in different years.

The amount of time a walk takes can vary among years but should be recorded. In some years, some species may be more apparent than in other years or species expression may occur in different areas, and thus species lists among years can be different. Monitoring should generally cover the same area and be done with relatively the same amount of intensity among years. However, areas heavily encroached by woody species may take less time to walk through after woody species thinning and prescribed fire has occurred. Conversely, increase in species richness with management may cause an associated increase in needed monitoring effort in some phases of savanna restoration. A GPS in tracking mode can be carried during the meandering walk to create a record of routes taken over time. Previous meandering walk routes can be useful in understanding where species richness data specifically originated, but the botanist is encouraged to intuitively search for plant species instead of attempting to follow former routes.

All species encountered whether native or non-native should be recorded along with general abundance and distribution within each soil moisture type. Example data sheets are in SM 1.

Data field details

OVERALL SITE: Record the name of the overall site.

MANAGEMENT UNIT: Record the name of the management unit (the larger savanna unit if there is one).

MONITORING AREA: Record the name of the savanna monitoring area. This can be a subunit of the larger management unit and is the area targeted by the meandering walk.

SOIL MOISTURE TYPE: Record “dry,” “mesic,” or “wet,” as defined by the soil drainage class(es) within the search area (**SOP 7**). If more than one category occurs in each monitoring area, provide the approximate percent of the area in each moisture category, and search each moisture category separately. Write a comment if what is indicated in the soil survey appears different from what is found on the ground.

DATE: Record the date of the survey as yyyy/mm/dd.

OBSERVER(S): Record each observer’s name.

START TIME: Record the time at the start of observations using a 24-hr clock (e.g., 14:32 = 2:32 pm).

END TIME: Record the time at the end of observations using a 24-hr clock (e.g., 17:02 = 5:02 pm).

TOTAL TIME: Record the total amount of time in hours and minutes spent conducting the meandering walk (end time – start time).

AREA SEARCHED: Record the approximate total area searched in hectares. This should coincide with the area of the delineated polygon(s) created on your map or with GIS.

SPECIES: During the search, record each native and non-native species encountered using current scientific name. Any abbreviations, codes, or names used should be referenced to a master list of corresponding scientific names and the authority (source of the scientific names, such as USDA Plants, Flora of North America, or ITIS) used. If you cannot identify a plant to species, collect or photograph a specimen for future identification, and note genus, if possible, in comments. If possible, flag an unknown plant species, and return later when diagnostic characteristics are apparent. Track unknown plants without enough information to distinguish from other species by coding them discretely until they can be properly identified. Record very small seedlings that cannot be identified with confidence as “unknown forb seedling”, “unknown grass seedling” “unknown sedge seedling” or “unknown woody seedling”.

ABUNDANCE: Record the general abundance of each species within each moisture category as “sparse,” “occasional,” “frequent,” or “very common” relative to the size of the site surveyed, according to the definitions below (**Table SOP 1.1**). Evaluate trees as a separate category. Though subjective, these indications can be useful, particularly to the site manager through time.

Table SOP 1.1. Definitions of species abundance categories for the meandering walk.

Abundance Category	Definition
Sparse	Seldom occurring and usually occurring as an isolated plant; few total individuals observed
Occasional	Scattered plants in low numbers; sometimes occur in small clumps
Frequent	Many plants observed; species is regularly seen but rarely becomes dominant
Very Common	Species occurring in large numbers; can be dominant

DISTRIBUTION: Record the general distribution of each herbaceous and woody species as “localized” or “widespread.” A localized species is found in one or a few discrete spots in the area searched. A widespread species is dispersed broadly throughout the area. While many localized species will also be sparse, a frequent species could possibly be localized (e.g., one solid patch of reed canarygrass). Conversely, individuals of a sparse species might be found in several separate places (i.e., widespread). Record GPS points or map locations for notable indicator species, for concentrations of prairie, savanna or woodland species or invasive species.

COMMENTS: Record any relevant comments.

SOP 2: Woody Species Meandering Walk

The meandering walk is used to establish baseline initial conditions and track the progress of a monitoring area by recording relative densities of woody species present and general conditions in the area in a given year and through time. Canopy density or light levels provide insights about the capacity to support understory flora and changes through time. The observer strives to record tree species abundance, oak tree health, light availability to the understory and general abundance of herbaceous vegetation (not by species) within polygons during each meandering walk. Prior to monitoring, polygons are developed based on differences in tree densities or other characteristics as indicated by aerial photographs, and on prior knowledge of the site.

Attributes measured

- General herbaceous abundance (sparse, occasional, frequent, very common)
- Tree and shrub abundance (sparse, occasional, frequent, very common)
- Tree canopy cover (0, 10, 25, 50, 75, 100)
- Oak condition (healthy, unhealthy)
- Duration of the survey (hours, minutes)
- Area surveyed (hectares)

Equipment and supplies

- Map showing monitoring area boundary and soil moisture type boundaries if applicable; remote imagery is used to create polygons
- Polygons derived from recent and previous monitoring year's remote imagery to delineate areas with distinctive characteristics (eg: tree densities, unique forms or aggregations of oaks or other types of trees)
- GPS unit with map, if used
- Laminated reference charts, codes and photos for data entry
- Survey protocol
- Datasheets or mobile device equipped for data entry
- Clipboard
- Pencils/pens
- Watch/clock
- Field references, such as field guides or cheat sheets (hard copy or digital)
- Hand lens
- Plastic bags/cooler and/or plant press for specimen collection
- Camera for photopoints and to photograph unknown plants and notable characteristics such as tree condition or erosional or natural features
- Metric measuring tapes
- Compass

Survey timing and schedule

Woody species meandering walks are typically done annually and based on phenological cues. A walk may be performed any time when trees are fully leafed-out, but observers should use the same phenological cue to guide timing of the walk among years (eg: when full leaf-out is

initially achieved in spring). Conduct the walk according to the phenological timing and frequency determined for the survey by confirming with the Site Survey Coordinator or by checking archived data for dates and frequency of sampling during previous monitoring visits.

Sources of error

A brief list of sources of error to consider is:

- Inconsistent application of survey methods
- Observer bias in estimations
- Plant misidentification

Establishment of sampling units

Each discrete monitoring area delineated by soil moisture type (if more than one moisture type exists within the overall area) is the sampling unit that should be searched. Use the map you created as well as a GPS unit, to search the area.

Care should be used when monitoring near the boundary of adjacent monitoring areas, because there may be a region of overlap (or a gap in application) of management in a given unit. If areas are highly interspersed with unique features (e.g., with bluff and lowland topography) or if management is applied in very patchy or complicated ways such that monitoring them separately would be impractical, they may be monitored as one management area and treated as a single sample unit. Management units often include a mosaic of brightly lit areas with few to no trees (eg: fully sunlit areas embedded as an opening within or on the edge of a wooded area) and moderately lit areas with dappled sunlight (eg: primarily with open grown oaks) characteristic of savannas, and more densely shaded woodlands.

Data collection procedures

Within each monitoring area, do a walk to evaluate tree relative abundance and distribution for the area within each tree density/cover polygon that was delineated on an aerial photograph prior to monitoring. (see Data field details). Different polygons may have different numbers of stopping points; data from the stops is intended to provide a snapshot of conditions in the year monitored and the number of stops is at the discretion of the Survey Coordinator but should be consistent among years. If management actions lead to notable differences in conditions in areas within a polygon and addition of points is desired, make note of changes in numbers and locations of points. The route of the woody species walk (SOP 2) should generally match the route of the botanist's walk (See SOP 1). This entails traversing all distinct vegetation zones, including parts of the site with different light levels (from canopy closure, or from openings along trails, roads, utility rights-of-way etc.) moisture types, topography, or other characteristics that could yield additional plant species. Isolated individuals or patches of invasive species of special management concern should be noted with GPS or other mapping techniques.

Light levels, tree densities and thus herbaceous expression may change dramatically through time. The biologist or manager may want to guide technicians to be sure to visit certain parts of the area known to be of interest. Special features or conditions such as rock outcroppings, streams/rivers, eroded areas, seeps, soil slumps, trash dumps, utility lines etc. should be noted and mapped with GPS or alternative mapping techniques in the first year observed. Thereafter,

watch for changes in condition (such as a deepening erosion incision) if they occur since the last woody species meandering walk.

Condition of woody species including trees, shrubs and vines should be observed and noted (see Data field details). In particular, note general oak condition or special characteristics such as growth pattern (broadly spreading or predominantly upward branching; shade pruned or dead lower limbs; cut stumps) of oaks. Take note of oak species that are most desirable or most vulnerable to disturbance.

The amount of time a walk takes can vary among years but should be recorded. In some years, some characteristics may be more apparent than in other years or herbaceous expression may occur in different areas, and thus observations among years can be different. This trend indicates progress toward recovery in savanna remnants that are virtually devoid of herbaceous vegetation as they begin to produce more biomass, and can indicate increasing ability to burn an area given the need for contiguous fine fuels. Monitoring should generally cover the same area and be done with relatively the same amount of intensity among years. However, areas heavily encroached by undesirable woody species may eventually take less time to walk through after woody species thinning and prescribed fire has occurred. Likewise, changes in tree density can result in changes in light levels and thus, in herbaceous understory vegetation expression so monitoring should be done at least after every management event or natural event that causes changes in woody species density and condition. A GPS in tracking mode can be carried during the meandering walk to create a record of routes taken over time. The woody species meandering walk should be conducted initially and at pre-determined time steps or whenever a management action within a polygon is taken that can change woody species and related understory composition. In the latter case, it may not be necessary to monitor all polygons, but only those affected by management, until the time step occurs to conduct SOP 2 in the entire area.

A simple user guide will be provided for electronic data entry using Survey123 or other application, and when available, it will be located in our [monitoring](#) ServCat site.

All species encountered whether native or non-native should be recorded along with general abundance and distribution within each soil moisture type. A sample blank datasheet and an example filled out datasheet are attached to this protocol (SM 1).

Data field details

DATE: Record the date of the survey as yyyy/mm/dd.

OBSERVER(S): Record each observer's name.

OVERALL SITE: Record the name of the overall site as it will appear in the Database

MANAGEMENT UNIT: Record the name of the management unit (the larger savanna unit if there is one).

MONITORING NAME: Record the name of the savanna monitoring area. This can be a subunit of the larger management unit and is the area targeted by the woody species meandering walk.

POLYGON NAME: Record the name of each polygon delineated from an aerial photograph that is being examined. There can be several polygons in each monitoring area. Note the tree characteristic(s) that define each polygon (general tree density, presence of open grown oaks, unidentifiable dense trees, or landmarks).

SOIL MOISTURE TYPE: Record “dry,” “mesic,” or “wet,” as defined by the soil drainage class(es) within the search area (**SOP 7**). If more than one category occurs in each area, provide the approximate percent of the area in each moisture category, and search each moisture category separately. Write a comment if what is indicated if the soil survey appears different from what is found on the ground.

START TIME: Record the time at the start of observations using a 24-hr clock (e.g., 14:32 = 2:32 pm).

END TIME: Record the time at the end of observations using a 24-hr clock (e.g., 17:02 = 5:02 pm).

TOTAL TIME: Record the total amount of time in hours and minutes spent conducting the meandering walk (end time – start time).

AREA SEARCHED: Record the approximate total area searched in hectares. This should coincide with the area of the delineated polygon(s) created on your map or with GIS.

HERBACEOUS ABUNDANCE: Record the general abundance of herbaceous vegetation within each moisture category and within each polygon as “sparse,” “occasional,” “frequent,” or “very common”, according to the definitions below (**Table SOP 2.1**). Evaluate trees as a separate category. Though subjective, these indications can be useful, particularly to the site manager through time.

Table SOP 2.1. Definitions of herbaceous vegetation abundance categories for the woody species meandering walk.

Abundance Category	Definition
Sparse	Isolated plants or small patches observed, but mostly there is unvegetated, bare ground
Occasional	Scattered plants or patches of vegetation occur; patches are mostly small; bare ground dominates
Frequent	Many plants are observed existing in discontinuous but often large patches throughout much of the area
Very Common	Vegetation occurs in large portions of the savanna and is generally contiguous; bare ground is generally rare or does not apparently exist

TREE ABUNDANCE: Record the general abundance of trees within each moisture category and within each polygon as “sparse,” “occasional,” “frequent,” or “very common” relative to the size of the site surveyed, according to the definitions below (**Table SOP 2.2**).

Table SOP 2.2. Definitions of tree and shrub abundance categories for the woody species meandering walk.

Abundance Category	Definition
Sparse	Seldom occurring or absent but usually occurring as an isolated tree/shrub or a few total individuals
Occasional	Scattered trees/shrubs in low numbers; sometimes occur in small clumps; light easily reaches most of the understory
Frequent	Many trees/shrubs observed and are fairly ubiquitous; can be evenly distributed or clumpy but moderate light reaches the understory in much of the area
Dense	Trees/shrubs occurring in large numbers; can be dominant, significantly impeding light reaching the understory

TREE SIZE: Record size of tree within each moisture category and within each polygon as “small”, “intermediate”, “large”, “mixed” relative to the size of the site surveyed, according to the definitions in **Table SOP 2.2**.

Table SOP 2.1. Definitions of tree size categories for the woody species meandering walk.

Size Category	Definition
Small	Most trees with trunk diameters between 2.5 and 10 cm (1 and 4 inches); dominated by young trees.
Intermediate	Trunk diameter mostly between 10 to 25 cm (4-10 inches)
Large	Most trees > 25 cm (10 inches) in diameter; dominated by mature trees.
Very Large	Trees with trunks diameters of 46 cm (18 inches)
Mixed	Tree diameters include a wide range of diameters; no size category dominant

SHRUB ABUNDANCE: Record the general abundance of shrubs within each moisture category and within each polygon as “sparse,” “occasional,” “frequent,” or “dense” relative to the size of the site surveyed, according to the definitions in **Table SOP 2.2**.

TREE CANOPY COVER: Record estimated canopy cover within each moisture category as (0, 10, 25, 50, 75, 100) using a laminated version (for field use) of **Figure SOP 2.1**. This can be done at the same interval as tree structural class and abundance determination.

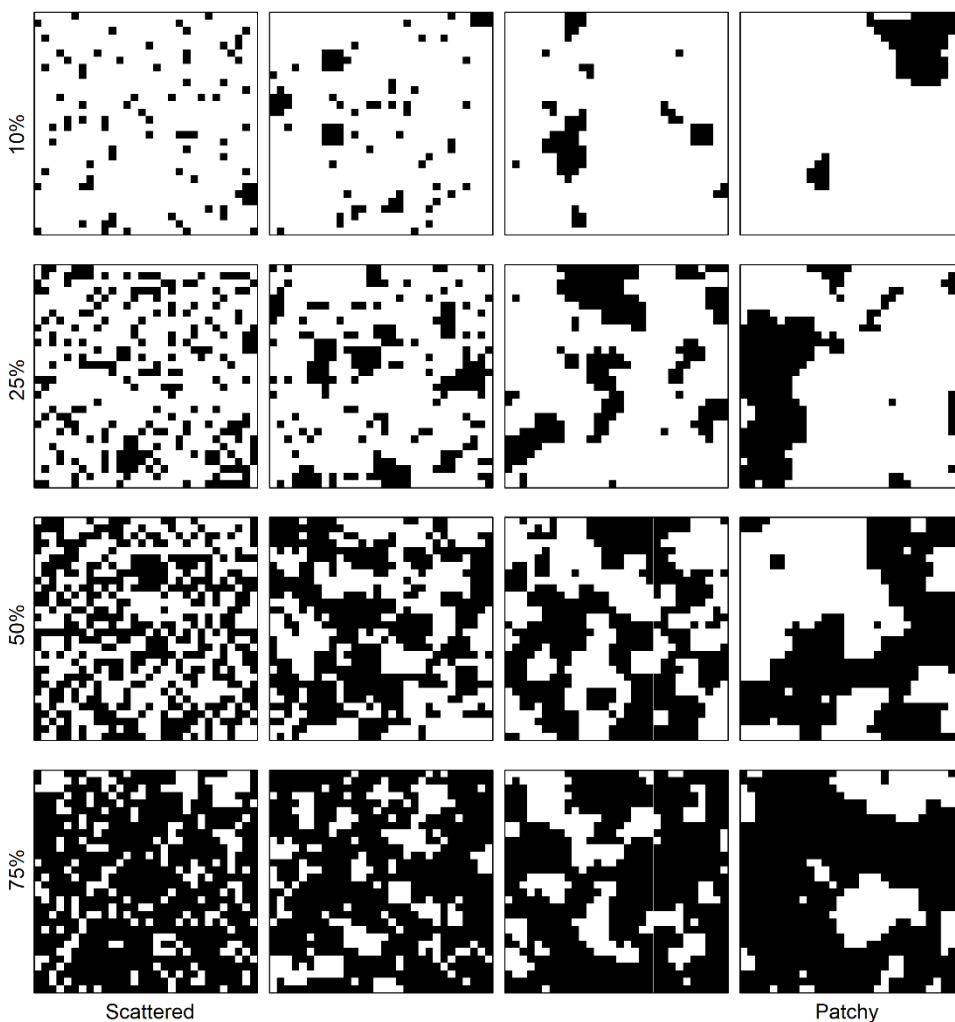


Figure SOP 2.1. Canopy cover chart used to guide estimation of canopy cover (0, 10, 25, 50, 75, 100 percent) including variations depending on degree of dispersion or patchiness.

OAK CONDITION: Observations on oak condition are important for evaluating tree health, potential longevity of overstory trees, wildlife habitat, restoration treatment effects and trends in tree condition. These observations are useful, for example, for making decisions about oak regeneration needs to sustain desired tree density and assessing adequacy and quality of wildlife habitat such as snags or den/cavity trees. During the initial meandering walk event, record the proportion of overstory oaks as healthy or unhealthy (**SOP Figure 2.2**), outward or upward branching form (**SOP Figure 2.3**) and dead trees by species. If species of dead tree cannot be determined, then identify as a member of either the white or black oak group. If this cannot be done, assign them to the dominant oak species. Record proportion of trees healthy versus

unhealthy to the nearest ten percent. Oak condition observations can be done at the same time as structural class determination. Additional overstory oaks can be opportunistically observed and recorded in the comments along the meandering walk between formal observations if desired.

Healthy trees may have:

- full, normal density crowns,
- fully expanded green leaves in summer

Healthy trees lack:

- major cavities
- massive limb breakage
- large or multiple fire or mechanical scars that are not recovering
- signs of decay of inner wood
- major dieback of branch tips in the upper crown.

Unhealthy or declining trees may have:

- less dense and abnormally small crowns,
- discolored leaves,
- dieback of major and terminal branches throughout and below the crown
- absence of sections of the crown due to stem or trunk breakage
- large cavities
- damage to or breakage of the main trunk

Note that healthy trees can have remnant dead branches within or at the base of the crown or along the trunk, but large dead horizontal branches can indicate intense shading or tree decline.

Frequent fire combined with dry conditions can result in small, gnarled oak trees with twisted trunks and asymmetric canopies characteristic of oak openings. This form can therefore be desirable and appropriate in savannas. In savannas experiencing frequent, hot fires, oak trees can be reduced to a bushy form originating from a basal caudex. This form, called a grub, was reported historically and is normal in some savanna types. In addition, savanna oaks can live for hundreds of years and many of these old oak trees will show signs of damage, recovery or tolerance of injury through time including scars that have partially or completely healed over and hollow trunks surrounded by healthy live tissue that supports the tree. These and other signs can be characteristic of a normal savanna tree. However, rapid or massive cumulative signs of disease or decline, large numbers of dead trees, or isolated incidences of disease caused by invasive organisms or adjacent land use are cause for concern.

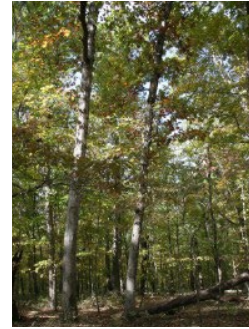


Figure SOP 2.2. Examples of healthy (top row) and unhealthy oak trees (bottom row).





Figure SOP 2.3. Oaks grown primarily in full or nearly full sunlight with outward branching pattern (top row) including some horizontal limbs and limb scars that occurred due to canopy closure and shading. Note that in oak openings, savanna oak trees may be diminutive and gnarled with stout trunks, twisted branching, and lopsided canopy. Trees growing densely and in high competition for light have tall straight boles free of large limbs along the lower trunk with canopies developed in the upper third of the tree generally in an upward branching pattern (bottom row).

OAK SEEDLING/SPROUT ABUNDANCE: Record abundance of seedlings or sprouts within each moisture category and within each polygon as “sparse,” “occasional,” “frequent,” or “very common” relative to the size of the site surveyed, according to the definitions below (**Table SOP 2.2**).

Table SOP 2.2. Definitions of oak seedling/sprout abundance categories for the woody species meandering walk.

Abundance Category	Definition
Sparse	Seldom occurring or absent
Occasional	Scattered seedlings or sprouts occur in low numbers or sometimes in small isolated patches
Frequent	Many seedlings/sprouts observed; can be evenly distributed or clumpy
Very Common	Seedlings/sprouts generally occur in large numbers throughout much of the area



Figure SOP 2.4. One year-old pin oak seedling (left), note acorn still attached and intact original stem; bur oak sprout (center and right), note multiple older, dead stems in contrast to two live sprouts that emerged in the current year

COMMENTS: Record any relevant comments including presence and GPS or map locations of special features such as rock outcroppings, streams/rivers, erosion, seeps, slumping soil, trash dumps, archaeological sites, buildings, utility lines etc. in the first year observed. Record changes in condition (such as a deepening erosion incision) if they occurred since the last meandering walk. Record comments about types of disease or decline in oaks or other savanna trees.

SOP 3: Tree Macroplots

In tree macroplots, the observer records living and dead (> 12.7 cm DBH) mature trees by species using the variable radius plot method. The macroplot also provides a geospatial framework within which herbaceous nested plots, sapling, shrub and vine, oak seedling and sprout microplots are located. Tree macroplots provide quantitative data for comparison of tree species composition, structure and general health across monitoring areas or over time to assess oak tree regeneration and recruitment into the overstory canopy layer. Tree species diversity and evenness is also calculated. Only trees greater than 12.7 cm in diameter are considered in the tree macroplots.

A Monitoring Macroplot Site Diagram ([Appendix A](#)) visually illustrates how SOPs 3-6 are spatially integrated. A Monitoring SOP Checklist ([Appendix B](#)) is a guide to conducting tasks in an efficient order in the field when SOPs 3-6 are used together and it helps minimize impact of trampling prior to data collection for more sensitive metrics like herbaceous vegetation.

Attributes measured

- Site character (photopoint)
- Canopy cover (spherical densiometer reading at N, E, S, W)
- Mature tree species richness
- Diameter at breast height (in cm at 1.37 m = breast height, or inches at 4.5 feet)
- Tree density (as m²/ha or ft²/ac)
- Crown condition (healthy, light decline, moderate decline, severe decline, dead)
- Tree form (outward or upward)

- Tree height to live crown ratio (<10, 10-25, 26-50, 51-75, 75-100 percent)
- Tree crown light exposure (codes 1-5)
- Basal scars scar aspect (N, NE, E, SE, S, SW, W, NW), type (fire or mechanical) and measurement (height, width, depth; measurement is optional)

Equipment and supplies

- Map showing savanna boundary, soil moisture type boundaries if applicable, and sampling point locations; remote imagery or topography is especially useful
- GPS unit preloaded with coordinates of sampling points or appropriate map if not using a GPS
- Survey protocol
- Metric = 2 BAF prism (each plot tree represents 2m²/ha of basal area); English = 10 BAF prism (each plot tree represents 10 ft²/ac of basal area)
- Logger's tape or other cloth measuring tape
- Laminated cover category chart
- Survey pins
- Clinometer
- Datasheets or mobile device equipped for data entry
- Clipboard
- Pencils/pens
- List of species observed in prior surveys (if any), if desired
- Field references, such as field guides or cheat sheets (hard copy or digital)
- Hand lens
- Compass
- Binoculars
- Clippers for accessing trees and for collecting unknowns (if roots are not required)
- Narrow hand trowel or gardener's knife for collecting unknowns with roots (from outside the plot)
- Small ruler for fire scar measurement
- Range finder to aid in determining tree distance
- Plastic bags/cooler and/or plant press for specimen collection
- Camera to photograph unknown trees and for photopoints
- Calculator

Survey timing and schedule

The macroplots should be monitored after any major restoration action such as tree thinning or after storms that significantly damage trees, and thereafter at least every 3-5 years. Monitoring should occur when leaves are fully expressed and before senescence, and using phenological cues to guide when monitoring happens each year.

Sources of error

A brief list of sources of error to consider is:

- Inconsistent application of survey methods
- Tree misidentification

- Missing species
- Inaccurately measuring tree diameter
- Inaccurately determining which trees are inside each macroplot
- Inaccurately evaluating crown condition
- Inaccurately measuring overstory crown cover
- Improperly using canopy cover cards

Establishment of sampling units

Each plot will be located using a GPS unit with preloaded coordinates. Use appropriate GPS protocols to navigate to the monumented center point of the plot. If using an alternative non-GPS method of locating points, indicate this in the comments field, and make sure your methods are well documented.

Care should be used when monitoring near the boundary of adjacent savanna areas, because there may be a region of overlap (or a gap in application) of management. If areas are highly interspersed (e.g., with bluff and lowland topography) or if management is applied in very patchy or complicated ways such that monitoring them separately would be impractical, they may be monitored together and treated as a single sample unit.

Data collection procedures

Using a handheld GPS, or map and compass, navigate to each monumented sample location. Conduct a tree inventory (of trees > 12.7 cm) using a BAF 2m prism to determine trees to be sampled and to estimate tree density. The monumented point is the center of the sampling area, not the viewer's eye. In other words, the prism is held over the monumented point and is rotated around it; it does not move with the body or head of the viewer. The prism should be held in a vertical position and looking through the flat side which is held at right angles to the line of sight when looking at a tree to determine if a tree is "in" or "out" of the sampling plot (**Figure SOP 3.1**). Observations are made on "in" trees only. Using the prism, determine "in" trees by rotating it within a 360° arc around the permanent plot center and sighting in on each tree.



"In" tree



"Out" tree



"Borderline" tree

Figure SOP 3.1. Using the prism to determine sample "in" trees in variable plot sampling. "In" trees are trees for which the trunk overlaps when looking through the prism. "Out" trees are trees for which the trunk does not touch when looking through the prism. "Borderline" trees are trees that touch but do not

overlap. (photos courtesy of: <https://www.uhcl.edu/environmental-institute/texas-envirothon/documents/study-guide/forestry-how-to-use-a-prism.pdf> Accessed on March 28, 2022)

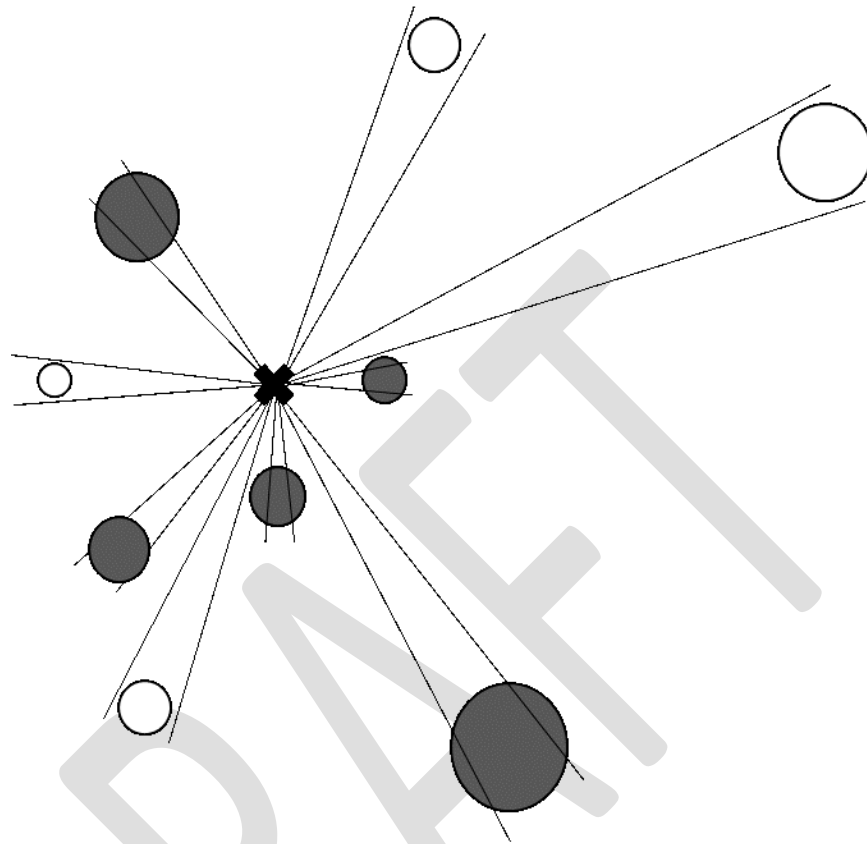


Figure SOP 3.2. Schematic of variable plot sampling. Shaded circles represent “in” trees and white circles represent “out” trees”

For trees that lean to the left or the right of the observer, if the tree appears “out” with an unrotated prism, then it is “out”. Otherwise, rotate the prism so that the vertical axis of the prism parallels the axis of the leaning tree, i.e., the sides of the prism should parallel the sides of the tree. Trees that lean severely toward or away from the observer and appear questionable should be checked by measuring the tree DBH (to nearest 0.1 cm) and horizontal distance to the face of the tree at the groundline (to nearest 0.01 meter). When vegetation obscures the observer’s line of sight to the tree, two options are available including 1) move the sample point so that the tree can be viewed (care must be taken to ensure the distance from the tree to the sample point remains the same) or 2) sight the tree at a visible point higher on the tree’s stem. In the second option, if the tree is “in” at this height, it is also “in” at breast height, unless the tree leans heavily toward the sampling point. If this is the case, or it appears slightly out or borderline, then it is best to measure the tree dbh and distance from sample point to the tree to determine if it is to be sampled (see how to do this below).

Borderline trees occur when a tree cannot be easily determined as in or out when using a prism. Borderline trees occur only when the distance from the sampling point to the face of the tree is equal to the DBH times plot radius factor (PRF). The PRF is determined based on the type of prism being used. The PRF for a BAF 2 prism is 0.3486 (meter per cm of DBH) for measurement from plot center to the face of the tree. See other PRFs for different BAF prisms in **Table SOP 3.1**.

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Table SOP 3.1. Plot radius factors for specified metric and English BAF prisms when sighting to the tree face from plot center.

Metric				English	
Basal Area Factor	Plot Radius Factor Tree Face	Basal Area Factor	Plot Radius Factor Tree Face	Basal Area Factor	Plot Radius Factor Tree Face
2	0.3486	13	0.1337	5	3.847
3	0.2837	14	0.1286	10	2.708
4	0.2450	15	0.1241	15	2.203
5	0.2186	16	0.1200	20	1.902
6	0.1991	18	0.1129	25	1.697
6.25	0.1950	20	0.1068	30	1.546
7	0.1840	20.25	0.1061	35	1.428
8	0.1718	24	0.0971	40	1.333
9	0.1617	25	0.0950	50	1.188
10	0.1531	30.25	0.0859	60	1.081
11	0.1458	32	0.0834		
12	0.1393	64	0.0575		

When a tree is borderline, a series of measurements and calculations must be made to determine if the tree is in or out. The horizontal distance from the sampling point to the face of the tree must be measured along with the DBH of the tree. The DBH should then be multiplied by the PRF, which will give a horizontal limiting distance (HLD). If the measured horizontal distance from plot center to the face of the tree is less than or equal to this HLD, then the tree is counted as in. If it is greater than the HLD, it is out.

Adjusting for slope

In point sampling, corrections must be made when the slope is $\geq 10\%$. Trees located on the contour with the sampling point need no adjustment for slope, they are in fact at 0% slope.

Adjustments for slope can be done in the field in one of two ways:

Method one – rotation of the prism (**Figure SOP 3.3**):

- If a tree on a slope appears “in” or “borderline” with an unrotated prism, it can be tallied as “in”, otherwise
- Measure the slope from the observer’s eye to tree at breast height with an instrument such as a clinometer
- Rotate the prism as if turning a steering wheel, an amount equal to the slope angle, keeping it perpendicular with the line of sight
- Sight the tree in question at breast height to determine if it is “in” or “out”

Method two - measurement and comparison to horizontal limiting distance:

- Measure the DBH to the 0.1 cm

- Determine the horizontal limiting distance from the face of the tree ($HLD = PRF \times DBH$); PRF is 0.3486 for a BAF 2 m prism when measuring to the face of the tree.
- Determine the percent of slope from the face of the tree at breast height to the sampling point using an instrument such as a clinometer.
- If the slope is 10% or greater, correct the horizontal limiting distance to slope limiting distance (SLD). To obtain the slope limiting distance, multiply the HLD by the appropriate slope correction factor (SCF) (**Table SOP 3.2**. ($SLD = HLD \times SCF$))
- Use a tape graduated in centimeters, decimeters, and meters to measure the distance from the face of the tree at breast height to plot center. The plot center is where the surveyors pin or wooden stake enters the ground. These are two exact points that can be measured "to" and "from". If the measured distance is equal to or less than the slope limiting distance, the tree is "in" and is sampled. If no slope correction is needed, the horizontal limiting distance is compared to the measured distance.

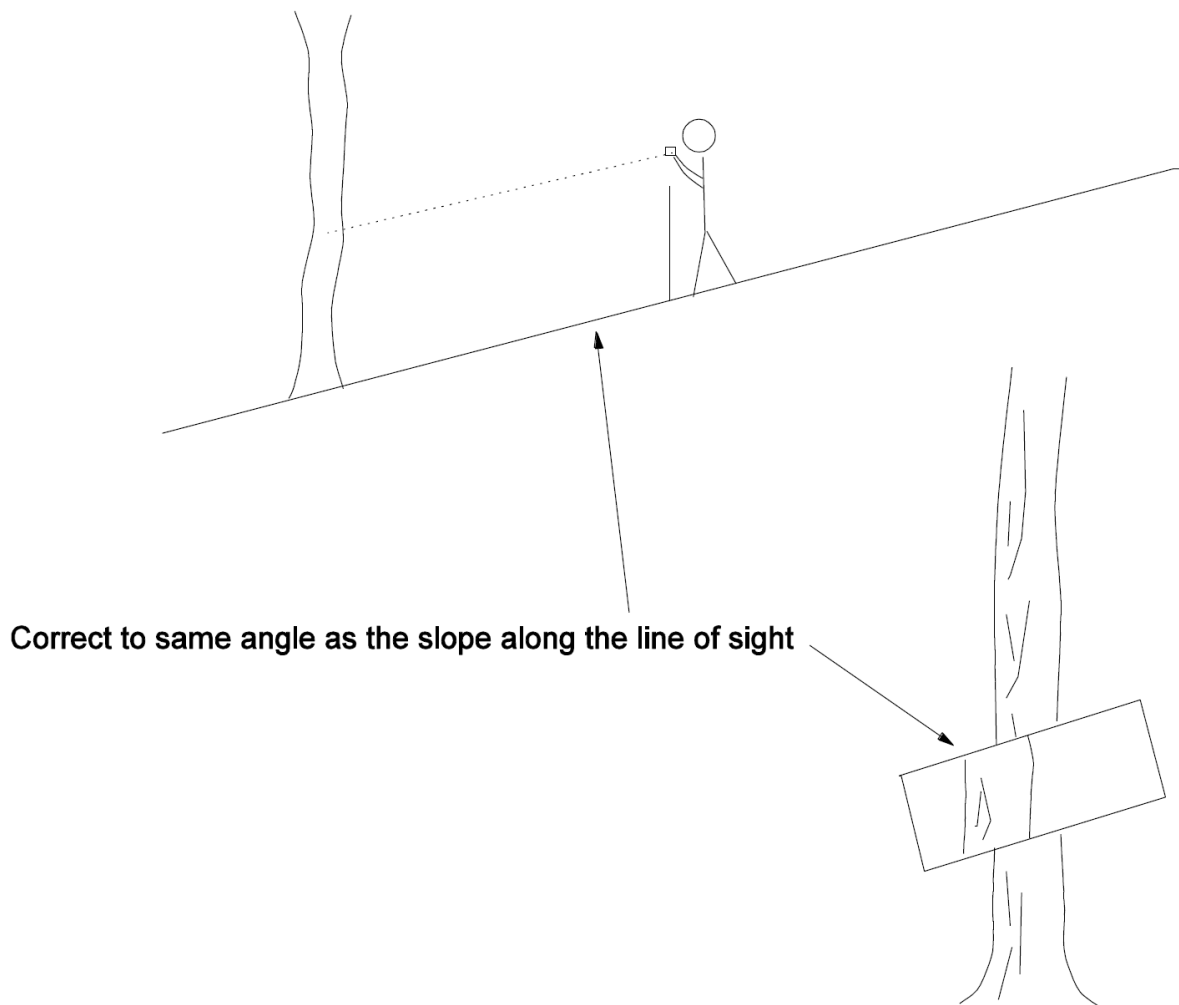


Figure SOP 3.3. Correcting for slope with the prism to identify sample trees (illustration from <https://www.uhcl.edu/environmental-institute/texas-envirothon/documents/study-guide/forestry-how-to-use-a-prism.pdf> accessed on March 28, 2022)

Table SOP 3.2. Slope correction factors for slopes of different degrees and percents.

Degrees slope	Percent slope	Correction factor	Degrees slope	Percent slope	Correction factor
0-4	0-9	1.00	34	68-69	1.21
5-9	10-17	1.01	35	70	1.22
10-12	18-22	1.02		71-72	1.23
13-14	23-26	1.03	36	73-74	1.24
15-16	27-30	1.04	37	75	1.25
17-18	31-33	1.05		76-77	1.26
19-20	34-36	1.06	38	78-79	1.27
21	37-39	1.07		80	1.28
22	40-42	1.08	39	81-82	1.29
23-24	43-44	1.09		83	1.30
25	45-47	1.10	40	84-85	1.31
26	48-49	1.11		86	1.32
27	50-51	1.12	41	87-88	1.33
28	52-53	1.13		89	1.34
29	54-55	1.14	42	90-91	1.35
30	56-57	1.15		92	1.36
	58-59	1.16	43	93-94	1.37
31	60-61	1.17		95	1.38
32	62-63	1.18	44	96-97	1.39
33	64-65	1.19		98	1.40
	66-67	1.20	45	99-100	1.41

Observations to make on all sample “in” trees:

For all “in” trees (≥ 12.7 cm DBH) record:

- Species
- DBH with diameter tape (to nearest 0.1 cm) of trees, 12.7 cm (**Figure SOP 3.4**)
- Tree crown condition (**Table SOP 3.3** for definitions)
- Tree live crown ratio (the ratio of the vertical length of live crown to the total tree height)
- Tree crown light exposure code (**Figures SOP 3.8; Table SOP 3.4** for definitions)
- Lower trunk fire scar height, width, and depth to nearest 0.1 cm, and percent of tree trunk circumference scarred using the greatest width of the scar (**Figure SOP 3.9**)

Locating DBH measurement point on a tree

On single-stemmed trees, DBH is measured 1.37 m above the groundline to the nearest 0.1 cm. For trees on slopes, DBH is measured on the uphill side of the tree. Common problems occur when measuring a tree’s DBH at or near a fork or swelling of the trunk. **Figure SOP 3.4** shows how to measure DBH when you encounter common abnormalities. For more complicated situations in measuring DBH, we refer you to the US Forest Service FIA field manual pp. 80-94 (https://www.fia.fs.fed.us/library/field-guides-methods-proc/docs/2019/core_ver9-0_10_2019_final_rev_2_10_2020.pdf accessed September 17, 2020).

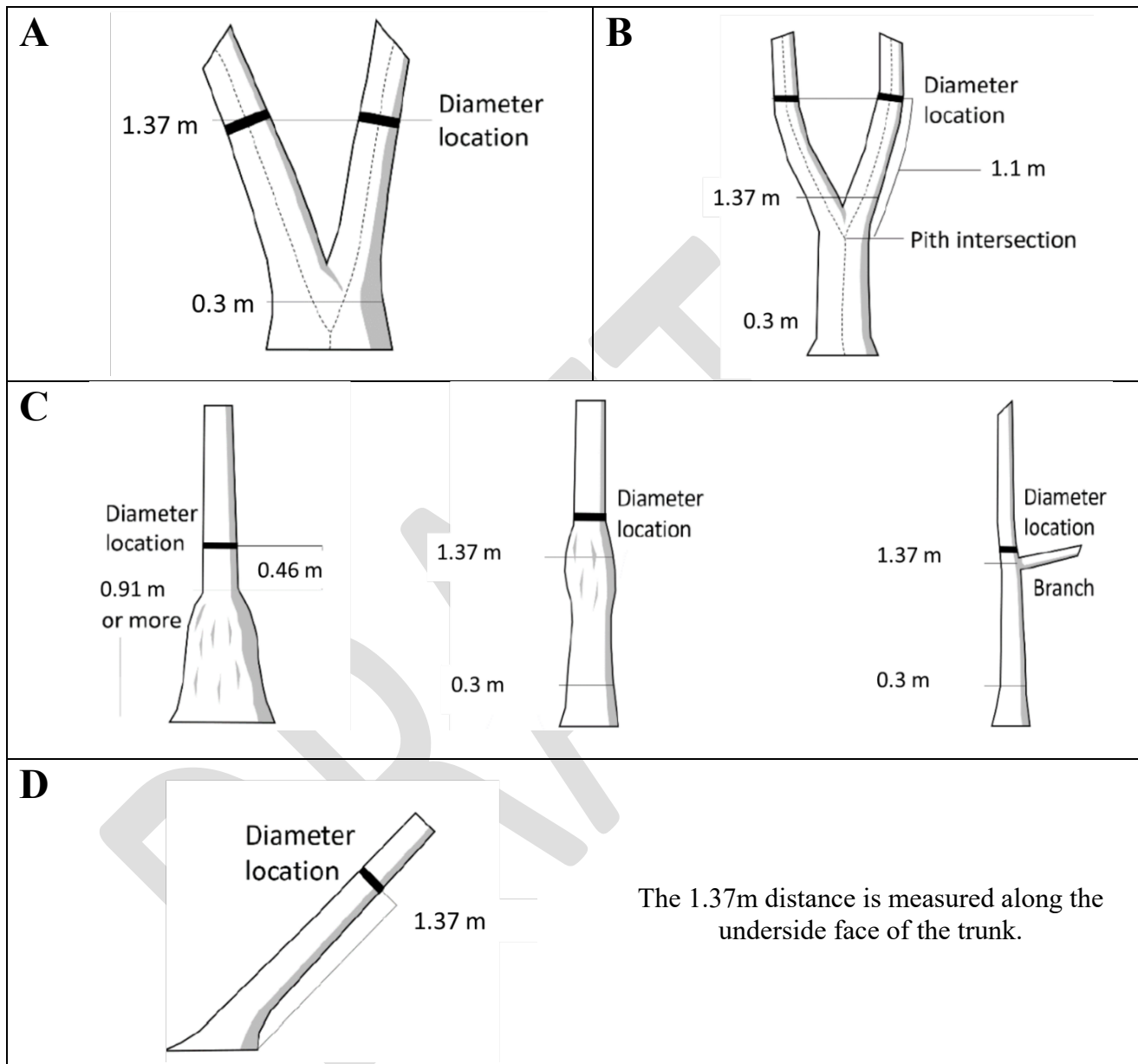


Figure SOP 3.4. Measuring a tree's DBH (diameter at breast height (1.37 m) with a diameter tape) under varying conditions. **A.** Trees forked below 0.3 m are treated as distinctly separate trees. DBH is measured for each stem at 1.37 m above the ground. When stems originate from pith (the center of the tree trunk) intersections below 0.3 m, it is possible for some stems to be within the limiting distance of the plot, and others to be beyond the limiting distance. Distances from plot center to the face of the tree are measured individually to the center of each stem where it splits from the stump. **B.** Trees forked between 0.3 m and 1.37 m are also counted as separate trees. The DBH of each fork is measured at a point 1.1 m above the pith intersection. When forks originate from pith intersections between 0.3 and 1.37 m, the limiting distance is the same for all forks--they are either all on, or all off the plot. **C.** Tree with butt-swell or bottleneck (left): Measure these trees 0.46 m above the end of the swell or bottleneck if the swell or

bottleneck extends 0.91 m or more above the ground. Trees with irregularities at DBH (middle and right): On trees with swellings, bumps, depressions, and branches at DBH, diameter will be measured immediately above the irregularity at the place it ceases to affect normal stem form. If the diameter point is out of reach above the irregularity, the diameter will be measured below the irregularity at the best repeatable location. If normal stem form does not exist, the diameter will be estimated at the prescribed location. **D. Leaning tree:** Measure diameter at the point that is 1.37 m from the base of the tree along the trunk.

Measuring DBH

Diameter of all “in” trees should be measured. To measure DBH, wrap a diameter tape around the tree at breast height, making sure the tape is level and not twisted. Read the diameter measurement where the tape overlaps with the zero marker. If the tree is leaning, then measure at an angle perpendicular to the axis of the tree. If the tree is forked, measure it as one tree if the fork occurs above DBH, or two trees if the fork is below DBH.

Crown Condition - Health

Observe crown condition from various vantage points and classify using categories in the table (Table SOP 3.3) below:

Table SOP 3.3. Tree crown condition classes.

Code	Condition	Definition
1	Healthy	< 10% of crown with branch or twig dieback or damage*, foliage discoloration, or dwarf leaves
2	Light decline	10 to 25% of crown with branch or twig dieback or damage, foliage discoloration, or dwarf leaves
3	Moderate decline	26 to 50% of crown with branch or twig dieback or damage, foliage discoloration, or dwarf leaves
4	Severe decline	Tree alive but > 50% of crown with branch or twig dieback or damage, foliage discoloration, or dwarf leaves
5	Dead	Standing dead tree, a snag

** crown dieback includes recent dead branches with fine twigs that occur in the outer portions of the tree crown. Dieback begins at the extremities of the crown and progresses toward the trunk until the entire branch is dead. Damage refers to broken branches or tree trunk resulting in loss of significant portions of the crown and major wounds that could accelerate wood decay development.*

Tree Form

Observe growth form of each oak tree and classify as generally “outward” or predominantly “upward” (**Figure SOP 3.5**) branching form. Trees with an outward branching pattern have grown in an open environment with high exposure to sunlight for most of their lives. They have large, lower, horizontal branches that sometimes occur relatively close to the ground and their crowns are often semicircular in profile. With intense shading from adjacent trees horizontal branches may die, break off, or eventually be reduced to stubs or knobs with bark growing over the wound. These features indicate that past light levels were sufficient to support a more open growth form. Trees with these characteristics are classified as having an “outward” branching form. The branches, stubs or knobs (**Figure SOP 3.6**) are noted as “shade-pruned”

Trees with an upward form have grown in dense competition with other trees and consistently exhibit a tall bole clear of major branches and a crown that is V-shaped (SOP 2.4).

Trees sometimes exhibit outward branching only on one side of the tree where light was abundant, but upward branching on a side that was heavily shaded by adjacent trees. When both outward and upward branching occurs on the same tree, they are included in the “outward” branching category.



Figure SOP 3.5. Oaks grown in abundant light in open stand conditions with an outward branching tree form (top row). Note horizontal branches and dead limbs or limb scars that indicate the tree grew in more open conditions in the past. Trees grown with intense competition for light (bottom row) have sharply upward branching tree form and a general absence of lower limbs.



Figure SOP 3.6. Oaks with dead branches, and stubs or knobs where branches once existed. Note rounded areas where bark has nearly completely obscured the site of a former limb.

Tree height to live crown ratio

Tree height to live crown ratio is the ratio of the vertical height of live crown to the total tree height. This does not include dead branches that exist above the live crown top. The base of the live crown is defined by the lowest live foliage that is included in the live crown. It does not include individual live branches with foliage that are < 2.5 cm in diameter that exist below and are disjointed from the live crown, nor any epicormic branches (fine stems, often in clusters, arising from the tree trunk or branches) below the live crown (**Figure SOP 3.7**). Record live crown ratio in the following percentage categories: <10, 10-25, 26-50, 51-75, 75-100.

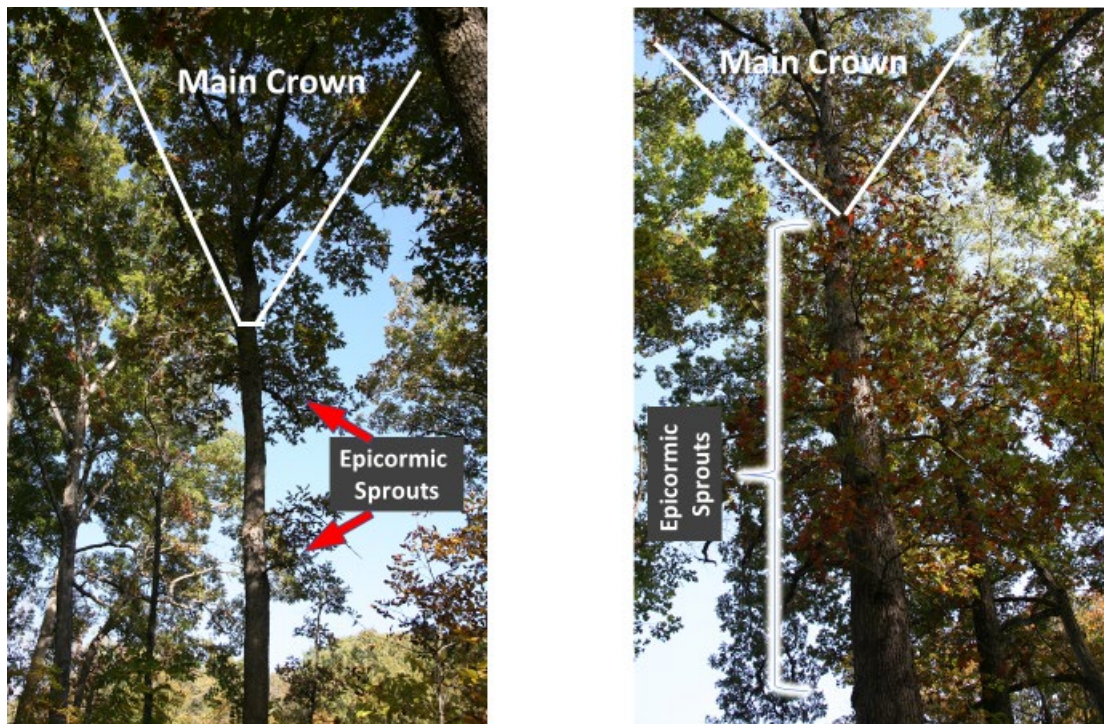


Figure SOP 3.7. Epicormic branches on a mature oak. They are often clusters, of fine branches arising from dormant buds under the tree bark. Often disjunct from the main tree crown, they may occur singly or in multiple sprouts along the tree trunk. They are not considered as part of the tree crown when determining tree height to live crown ratio.

Crown light exposure

Crown light exposure measures the amount of direct sunlight a live tree receives when the sun is directly overhead. Determining crown light exposure involves dividing the tree crown into five sections: four equal vertical quarters, i.e., faces, and the top (**Figure SOP 3.8**). Crowns are rated from zero to five, depending on the number of crown sections exposed to direct sunlight, considering the sun is directly overhead (**Table SOP 2.6**). Visually divide the crown into quarters, maximizing the number of quarters that are fully in direct sunlight. Each quarter should have at least a third of the live crown ratio exposed to direct sunlight throughout that section to count. Add one if the tree receives any direct light from the top. Add up the total number of sections exposed to full light, including the top if it receives any light, and record the total score. Record crown light exposure code (**Table SOP 3.4**) for each live tree. For example, if a tree is completely in direct sunlight on all sides with more than a third of live crown ratio receiving light, the tree would have a score of 5. Trees completely overtopped by their neighbors' crowns would have a score of 0. An intermediate situation is given in **Figure SOP 3.8**.

Visibility and viewing angle of the crown can cause evaluation problems. In dense stands it can be difficult to divide the crown accurately into quarters or see the crown top. Trying to establish live crown ratios for each quarter may be a challenge and make it hard to determine whether each entire quarter is receiving light. Do your best to make these determinations.

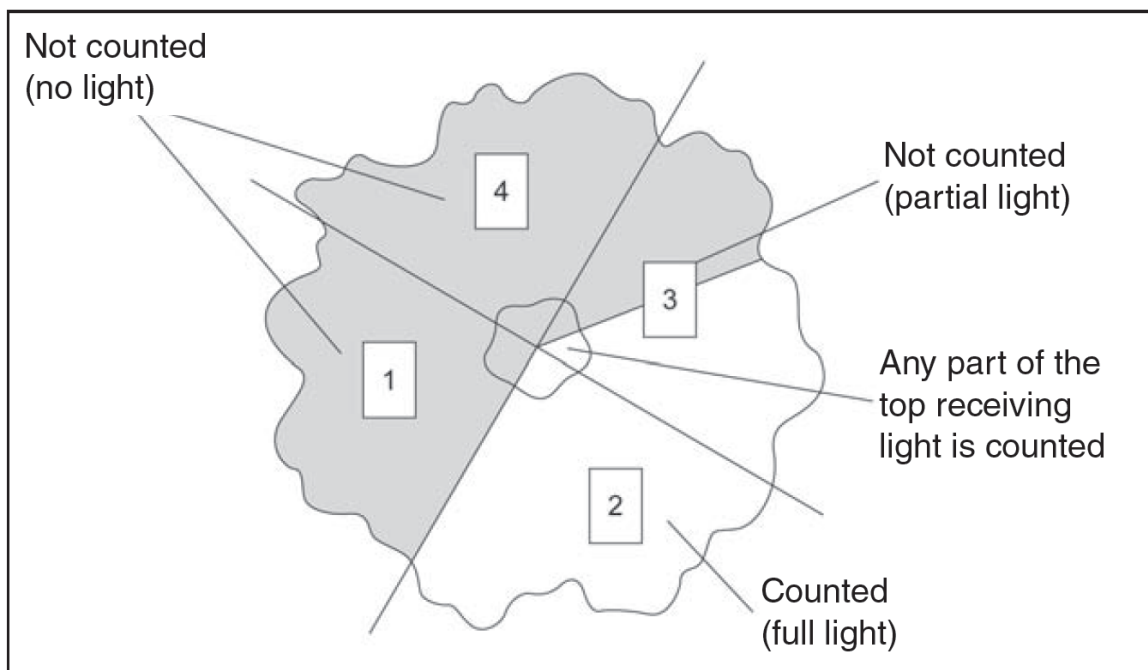


Figure SOP 3.8. When rating crown exposure, count only those sides that completely receive full light and have more than a third live crown ratio. If any portion of the top receives light, then it is counted as 1. In this example, the tree would be scored a 2.

Table SOP 3.4. Crown light exposure definitions.

Code	Definition
0	The tree receives no full light because it is shaded by trees, vines, or other vegetation
1	The tree receives full light from the top or 1 side
2	The tree receives full light from the top and 1 side (or 2 sides without the top)
3	The tree receives full light from the top and 2 sides (or 3 sides without the top)
4	The tree receives full light from the top and 3 sides
5	The tree receives full light from the top and 4 sides

Basal Scars

Fire scars at the base of oak trees result when a fire moving through a woodland is hot enough for enough time to cause damage to the bark and sometimes to penetrate more deeply into the cambium. These fire scars, known as “cat faces”, are usually triangular with a wide portion near the soil and an apex directly above it. Fire scars are often charred, or blackened by the fire that created the scar, or by subsequent fires that have kept the wound open as the cambium is repeatedly killed. Scar tissue is often evident and, in some cases, is smooth and has thinner bark compared to uninjured parts of the tree trunk. Some fire scars may be completely closed and covered by bark and woundwood, but the wound is still obvious.

Wounds on the lower trunk can be caused by mechanical damage from other trees falling into the standing tree, or from mowing, tree thinning or other management operations. Except for mowing, these wounds are generally higher on the trunk, do not always contact the ground, and

are often oval in shape compared to triangular catfaces resulting from fire. Mowing wounds are often horizontal to the ground, and linear with evidence of torn bark.



Figure SOP 3.9. Fire scars on mature oak trees (A) encircled area shows the scar with live bark recovering on a buttress root (B) large open scar from serious injury and repeated frequent burning.

Count and measure (measurement is optional) all fire scars originating at or near the base of the trunk of each tree (**Figure SOP 3.9**). Height (H) is from the ground line to the highest extent of the scar. Width (W) is distance of the widest portion of the scar, normally the base of the scar. Tree circumference is measured at the widest part of the scar. Scar depth (D) measures the extent of the scar into the tree from the exterior surface to the innermost surface. Record the orientation (N, NE, E, SE, S, SW, W, NW) of the fire scar on the tree. Mechanical scars can be noted, but recording and measuring is optional and depends on management objectives.

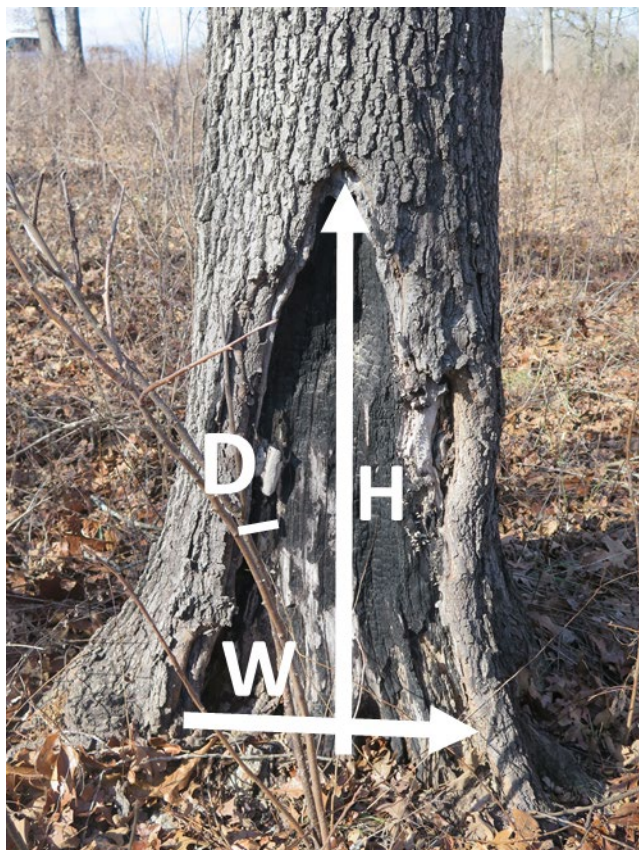


Figure SOP 3.10. Measuring a fire scar width (W), height (H), and depth (D).

On each macroplot, determine percent canopy cover at plot center using a concave spherical densiometer. Make 4 observations by facing in the cardinal directions about plot center. Use the center as a photopoint to also take photographs in 4 cardinal directions at eye level.

A simple user guide will be provided for electronic data entry using Survey123 or other application, and when available, it will be located in our [monitoring](#) ServCat site.

A sample blank datasheet and an example filled out datasheet are attached to this protocol (Supplemental Materials 1)

Data field details

OVERALL SITE: Record the name of the overall site.

MANAGEMENT UNIT: Record the name of the management unit (the larger savanna unit if there is one).

MONITORING AREA: Record the name of the savanna monitoring area. This can be a subunit of the larger management unit and is the area targeted by the tree macroplots.

SOIL MOISTURE TYPE: Record “dry,” “mesic,” or “wet,” as defined by the soil drainage class(es) within the search area (SOP 7). Write a comment if what is indicated in the soil survey appears different from what is found on the ground.

DATE: Record the date of the survey as yyyy/mm/dd.

OBSERVER(S): Record each observer’s name.

MACROPLOT NUMBER: Record the macroplot number. Assign a sequential number for each macroplot established.

MACROPLOT GPS: Record the GPS coordinates for the macroplot center or use other method to geospatially record location.

ASPECT: Record macroplot aspect (0-360 azimuth).

SLOPE: Record percent slope of macroplot.

PHOTOPOINTS: Using a cell phone camera or other digital camera take a photo at eye level in each cardinal direction (N, E, S, W) from plot center such that all 4 photos provide a 360° view of the macroplot. Label photo file identifying the site name, savanna unit, microplot number and date of photo or attach to the site using the electronic application to be developed in the future

CANOPY COVER: Record percent canopy cover using a spherical densitometer at the center of each plot in each of the four cardinal directions (N, E, S, W).

TREE NUMBER: Assign a sequential number (1, 2, 3, 4...) to each tree sampled. This number should be used for any measurement assigned to that tree. It applies only to the current survey and the tree need not be tagged for future reference.

SPECIES: For each tree determined to be “in” based on the 2m BAF prism, record species. Record each native and non-native species encountered using current scientific name. Any abbreviations, codes, or names used should be referenced to a master list of corresponding scientific names and the authority (source of the scientific names, such as USDA Plants, Flora of North America, or ITIS) used. If you cannot identify a plant to species, collect or photograph a specimen for future identification, and note genus, if possible, in comments. If possible, flag an unknown tree species, and return later when diagnostic characteristics are apparent. For plants with enough information to distinguish from other species, keep track of unknowns by coding them discretely until they can be properly identified.

DBH: Record the diameter of each “in” tree to the nearest 0.1 cm using a diameter tape located at 1.37 m from the ground. Use instructions in **Figure SOP 3.4** to understand how to measure DBH if trees are on a slope or have abnormal forms.

TREE CROWN CONDITION: Record codes 1-5 (healthy, light decline, moderate decline, severe decline, or dead) based on definitions found in **Table SOP 3.3**

TREE FORM: Record “O” for outward form and “U” for upward growth form for each “in” tree (**Figure SOP 3.5**).

TREE HEIGHT TO LIVE CROWN RATIO: Record tree live crown ratio in the following percentage classes: <10, 10-25, 26-50, 51-75, 75-100 for all “in” trees.

TREE CROWN LIGHT: Record tree crown light exposure using codes 0-5 corresponding to conditions expressed in **Table SOP 3.4** and **Figure SOP 3.8**.

BASAL SCAR: Record height, width and depth of each fire scar (**Figure SOP 3.10**) on each tree to the nearest 0.5 cm (optional). Record type of scar (fire or other). Record tree circumference at the widest part of the scar. Record quadrant that scar mostly occupies using cardinal directions (East, South, North, West, Northeast, Southeast, Southwest, Northwest).

COMMENTS: Record any relevant comments, including presence of a mowed or disturbed area, citing the reason for disturbance if known.

SOP 4: Nested Frequency Plots

In nested frequency plots, the observer uses a frame to assess the frequency of plant species in a monitoring area. The nested frequency plots provide quantitative data for comparison across management areas or over time, as well as for calculation of diversity and evenness.

A Monitoring Macroplot Site Diagram (Appendix A) visually illustrates how SOPs 3-6 are spatially integrated. A Monitoring SOP Checklist (Appendix B) is a guide to conducting tasks in an efficient order in the field when SOPs 3-6 are used together and it helps minimize impact of trampling prior to data collection for more sensitive metrics like herbaceous vegetation.

Attributes measured

- Species encountered in each plot (frequency of occurrence)
- Smallest subplot in which each species occurred

Equipment and supplies

- Map showing monitoring area boundary, soil moisture type boundaries if applicable, and sampling point locations; satellite imagery or topography is helpful
- GPS unit preloaded with coordinates of sampling points or appropriate map if not using a GPS
- Survey protocol
- Plot frame
- Datasheets or mobile device equipped for data entry
- Clipboard
- Pencils/pens
- List of species planted, if desired
- List of species observed in prior surveys (if any), if desired
- Field references, such as field guides or cheat sheets (hard copy or digital)

- Hand lens
- Compass
- Plastic bags/cooler and/or plant press for specimen collection
- Camera to photograph unknown plants
- Kneepads
- Clippers to remove nearby woodies to position the frame. Note: plants within the frame should not be removed. Also useful in taking samples of unknowns when roots are not required.
- Gardeners knife for digging an unknown species when roots are necessary. Note: only dig plants located outside of the plot frame.
- Flags to mark nested plots
- Gloves

Survey timing and schedule

The nested frequency plots should be implemented initially and at least every 3–5 years after the initial restoration management phase and based on phenological cues.

Sources of error

A brief list of sources of error to consider is:

- Inconsistent application of survey methods
- Plant misidentification
- Missing species
- Inaccurately determining which plants are inside each subplot

Establishment of sampling units

Each plot will be located using a GPS unit with preloaded coordinates or map with marked points that correspond to Tree Macroplot monuments (see **SOP 3**). Use appropriate GPS protocols to navigate to the monumented point. When the point is located, move 4-m north of the point to establish the sample location of the herbaceous nested frequency plot frame to minimize trampling that could occur during woody species sampling. If using an alternative non-GPS method of locating points (Element 3), indicate this in the comments field, and make sure your methods are well documented. Follow guidance in Element 3 about where to avoid sampling.

Care should be used when monitoring near the boundary of adjacent monitoring areas, because there may be a region of overlap (or a gap in application) of management in a given unit. If monitoring areas are highly interspersed (e.g., in ridge and swale topography) or if management is applied in very patchy or complicated ways such that monitoring them separately would be impractical, they may be monitored as one monitoring area and treated as a single sample unit.

Data collection procedures

Using a handheld GPS, or map and compass, navigate to each sample location. Orient the plot frame consistently for all plots. For example, facing north, place the plot frame in front of and to the left of the observer with subplot A (the smallest subplot) always farthest to the left and closer to the observer; **Figure SOP 4.1**). Alternatively, a random compass direction can be chosen prior to going to the field. To prevent bias when the sample point has been approached, the orientation of the plot frame must be predetermined. Do a minimum of 4 nested frequency plots, one in each

cardinal direction. If the macroplot is large enough, increase the number of nested frequency plots to 8 and locate them between each cardinal direction.

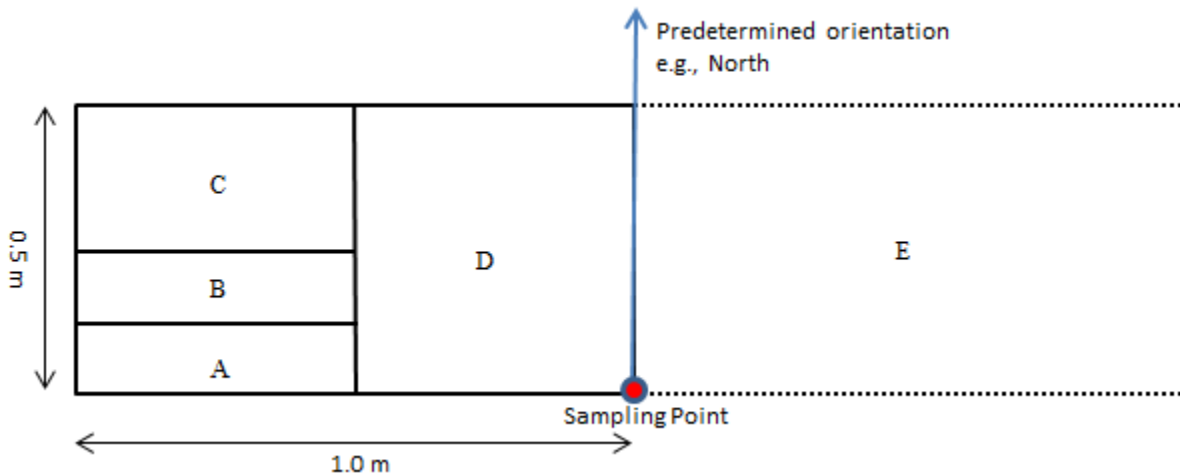


Figure SOP 4.1. Nested plot frame with example predetermined orientation.

Carefully adjust the vegetation so that only plants rooted within each subplot division are recorded. It may be desirable to remove one end of the frame to more easily slide it into tall vegetation. Replace the end to allow accurate monitoring. Record all native and non-native species, blooming and non-blooming, in the smallest subplot, and record an A for the appropriate plot on the datasheet or electronic application. Continue inspecting subplots in order from smallest to largest and record new species along with the subplot in which they first occurred. For any given 1-m² plot, a species will only be recorded once and only in the smallest subplot in which it occurred. The final subplot (E) is always the 50 cm x 100 cm area of the plot frame after it is “flipped” on its short axis as indicated in **Figure SOP 4.1**. If using a paper datasheet, use a separate column for each plot; adding datasheets as necessary to accommodate number of plots needed. Also use additional datasheets to accommodate more than 28 species. A simple user guide will be provided for electronic data entry using Survey123 or other application, and when available, it will be located in our [monitoring](#) ServCat site.

A sample blank datasheet and an example filled out datasheet are attached to this protocol ([Appendix](#)). The key thing to note is that each species is entered only once at each sample plot location, and the smallest subplot within which it is found is entered under the appropriate plot number.

Note the nested plot sizes on the datasheet or electronic application and in the Database. Ensure the same sampling subplots are used for each monitoring event.

Data field details

OVERALL SITE: Record the name of the overall site.

MANAGEMENT UNIT: Record the name of the management unit (the larger savanna unit if there is one).

MONITORING AREA: Record the name of the savanna monitoring area. This can be a subunit of the larger management unit and is the area targeted by the nested frequency plots.

SOIL MOISTURE TYPE: Record “dry,” “mesic,” or “wet,” as defined by the soil moisture categories within the monitoring area (SOP 3).

DATE: Record the date of the survey as yyyy/mm/dd.

OBSERVER(S): Record each observer’s name.

NESTED PLOT SIZES: Dimensions in cm are given for nested plot sizes 1–5 (**Table 3.2**).

MACROPLOT NUMBER: Record the macroplot number.

GPS MACROPLOT: Record GPS coordinates of the macroplot.

GPS NESTED PLOT: Record the GPS coordinates of the nested plot location.

SOIL MOISTURE TYPE PER POINT: Document the soil moisture type (SOP 7) for each sampling point (plot number). Write a comment if what is indicated in the soil survey appears different from what is found on the ground.

SPECIES: Record each native and non-native species encountered using current scientific name. Any abbreviations, codes, or names used should be referenced to a master list of corresponding scientific names and the authority (source of the scientific names, such as USDA Plants, Flora of North America, or ITIS) used. If you cannot identify a plant to species, collect or photograph a specimen for future identification (Appendix E), and note genus, if possible, in comments. If possible, flag an unknown plant species, and return later when diagnostic characteristics are apparent. For plants with enough information to distinguish from other species, keep track of unknowns by coding them discretely until they can be properly identified. Record very small seedlings that cannot be identified with confidence as either “unknown forb seedling” or “unknown grass seedling” (Element 2).

SMALLEST SUBPLOT: For each species, record the smallest subplot (A, B, C, D, or E) in which that species occurred for each plot, as instructed in Data Collection Procedures. On paper datasheets, if a species did not occur in a given plot, leave the cell blank, or strike through it.

COMMENTS: Record any relevant comments, including presence of a mowed or disturbed area, citing the reason for disturbance if known.

SOP 5: Tree Sapling, Shrub and Vine Microplot

Sapling, Shrub and Vine Microplots are used to record counts, size classes and cover of saplings, shrubs and all vines. Tree resprouts are recorded if they are in the appropriate size class. These plots provide quantitative data for comparison of the composition and structure of immature trees and other woody species that occur across monitoring areas or over time to assess encroachment

and in the understory of trees or in more open areas. Tree sapling, shrub and vine species diversity and evenness are also calculated.

If sufficient monitoring staff is available, it can be helpful to divide the microplot in half, assigning one observer to each half, and a third person located in the center to record data called out from each person. Care must be taken that no overlap occurs in the center of the plot if this method is used.

A Monitoring Macroplot Site Diagram (Appendix A) visually illustrates how SOPs 3-6 are spatially integrated. A Monitoring SOP Checklist (Appendix B) is a guide to conducting tasks in an efficient order in the field when SOPs 3-6 are used together and it helps minimize impact of trampling prior to data collection for more sensitive metrics like herbaceous vegetation.

Attributes measured

- Species
- Count of saplings
- Crown cover by percent of shrubs and vines
- Average shrub height by species,

Equipment and supplies

- Map showing savanna boundary, soil moisture type boundaries if applicable, and sampling point locations; satellite imagery or topography is helpful
- GPS unit preloaded with coordinates of sampling points or appropriate map if not using a GPS
- Survey protocol
- Datasheets or mobile device equipped for data entry
- Clipboard
- Pencils/pens
- List of species observed in prior surveys (if any), if desired
- Field references, such as field guides or cheat sheets (hard copy or digital)
- 4-foot fluorescent flags attached to 4-foot bamboo stakes and/or fluorescent flagging tape
- Hand lens
- Compass
- Metric cloth measuring tape and diameter tape
- Small ruler to measure saplings and shrubs
- Clippers to access trees for measuring
- Surveyor's pins
- Plastic bags/cooler and/or plant press for specimen collection
- Camera to photograph unknown woody species

Survey timing and schedule

The sapling, shrub and vine microplots should be implemented after any major restoration action such as tree thinning and thereafter at least every 3-5 years.

Sources of error

See Element 2 for a full description of sources of error. A brief list to consider is:

- Inconsistent application of survey methods
- Species misidentification
- Missing species

Establishment of sampling units

Each plot will be located using a GPS unit with preloaded coordinates. Use appropriate GPS protocols to navigate to the monumented center point of the tree macroplot which also serves as the center of the 7.3 m radius tree sapling, shrub and vine microplot. If using an alternative non-GPS method of locating points indicate this in the comments field, and make sure your methods are well documented.

Care should be used when monitoring near the boundary of adjacent savanna areas, because there may be a region of overlap (or a gap in application) of management in a given unit. If savanna areas are highly interspersed (e.g., with bluff and lowland topography) or if management is applied in very patchy or complicated ways such that monitoring them separately would be impractical, they may be monitored as one monitoring area and treated as a single sample unit.

Data collection procedures

Using a handheld GPS, or map and compass, navigate to each monumented sample location. The center of the sapling, shrub and vine microplot is at the center of the tree macroplot. Establish one 7.3 m radius and visually define a circular sampling plot with flags or flagging tape. Conduct the following sapling, shrub and vine survey:

- For all live tree saplings (≥ 2.5 cm DBH but < 12.7 cm DBH) record:
 - Species
 - Stem count by species in two DBH classes including 2.5 to 7.6 cm and 7.6 to 12.7 cm; if count exceeds 25 stems, cease counting and record as “m” (for > 25 stems)
- For all shrubs record:
 - Species
 - Percent cover estimates by category (0, 10, 25, 50, 75, 100) (**Figure SOP 5.1**)
 - Average height of shrub canopy by species
 - A shrub is a woody plant that is typically less than 25 feet tall at maturity. Unlike a tree, shrubs have multiple stems, lack a well-defined main stem, and vary widely in size; some shrubs are less than 6 feet high while others are around 20 to 25 feet tall
- For all vines (herbaceous and woody) record:
 - Species
 - Percent cover estimates by category (0, 10, 25, 50, 75, 100) (**Figure SOP 5.1**)

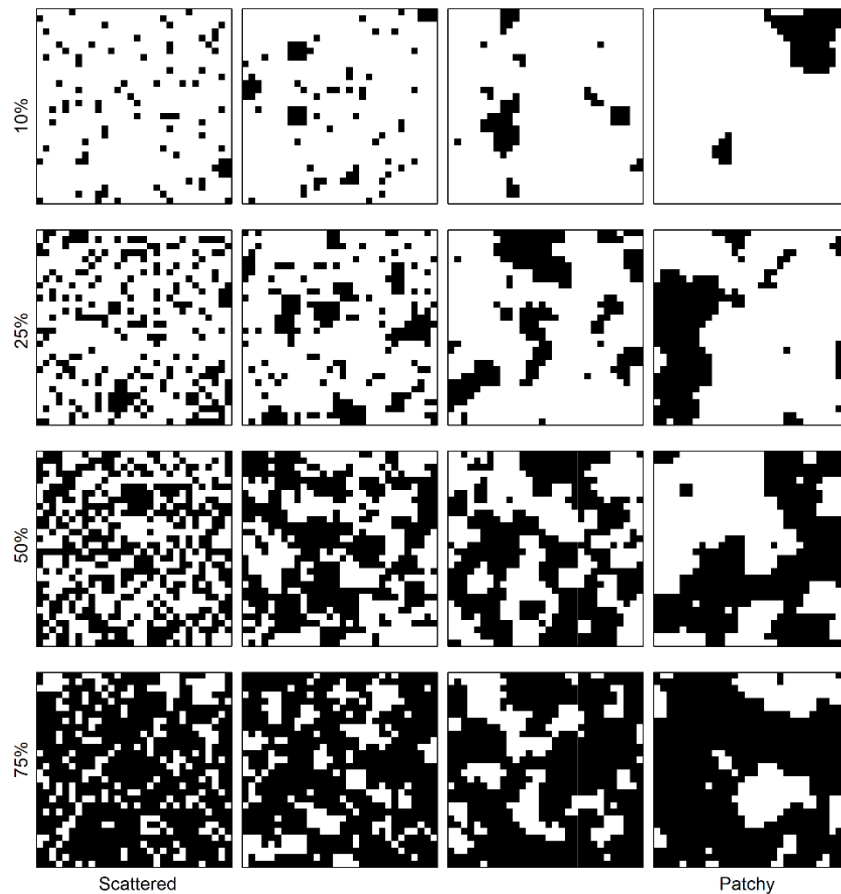


Figure SOP 5.1. Percent cover chart used to guide estimating canopy cover (0, 10, 25, 50, 75, 100) including variations depending on degree of dispersion or patchiness.

A simple user guide will be provided for electronic data entry using Survey123 or other application, and when available, it will be located in our [monitoring](#) ServCat site.

A sample blank datasheet and an example filled out datasheet are attached to this protocol ([Appendix](#)).

Data field details

OVERALL SITE: Record the name of the overall site.

MANAGEMENT UNIT: Record the name of the management unit (the larger savanna unit if there is one).

MONITORING AREA: Record the name of the savanna monitoring area. This can be a subunit of the larger management unit and is the area targeted by the tree sapling, shrub and vine microplots.

SOIL MOISTURE TYPE: Record “dry,” “mesic,” or “wet,” as defined by the soil moisture categories within the monitoring area (SOP 7).

DATE: Record the date of the survey as yyyy/mm/dd.

OBSERVER(S): Record each observer’s name.

MACROPLOT NUMBER: Record the macroplot number.

MICROPLOT NUMBER: Assign a unique sequential number (1, 2, 3, 4...) to each Tree Sapling, Shrub and Vine Microplot.

SPECIES: For trees saplings ≥ 2.5 cm DBH but < 12.7 cm DBH, and shrubs, and vines on microplots, record species using current scientific name in appropriate categories. Any abbreviations, codes or names used should be referenced to a master list of corresponding scientific names and the authority (source of the scientific names such as USDA Plants, Flora of North America, or ITIS) used. If you cannot identify a plant to species, record genus if possible and/or collect and/or photograph a specimen for future identification. For plants with enough information to distinguish from other species, keep track of unknowns by coding them discretely until they can be properly identified. Record very small plants that cannot be identified with confidence as either “unknown sapling”, “unknown shrub seedling” or “unknown vine seedling”.

STEM COUNTS: Record counts of sapling stems by species in two DBH classes including 2.5 to 7.6 cm and 7.7 to 12.7 cm

SHRUB PERCENT COVER: Visually estimate and record percent crown cover of shrubs and vines by species (**Figure SOP 5.1**) in the appropriate cover class (0, 10, 25, 50, 75, 100).

SHRUB HEIGHT: Record average height of shrub canopy by species to nearest 0.5 m.

VINE PERCENT COVER: Record percent cover by the appropriate cover class (0, 10, 25, 50, 75, 100).

COMMENTS: Record any relevant comments, such as presence of a mowed or disturbed area, citing the reason for disturbance if known, or concentrations of highly invasive species

SOP 6: Oak Seedling & Sprout Microplots

Oak seedling and sprout microplots are used to record number of oak seedlings and sprouts by species in each microplot. A seedling is a young tree that has recently germinated from an acorn. Often one can discover an acorn still attached to the seedling. A sprout is a small tree with multiple stems that sprout after the original top growth is killed by fire or clipping. Oaks can

persist as a sprout for many years. Oak seedling & sprout microplots provide quantitative data to understand oak regeneration, seedling diversity and evenness.

A Monitoring Macroplot Site Diagram (Appendix A) visually illustrates how SOPs 3-6 are spatially integrated. A Monitoring SOP Checklist (Appendix B) is a guide to conducting tasks in an efficient order in the field when SOPs 3-6 are used together and it helps minimize impact of trampling prior to data collection for more sensitive metrics like herbaceous vegetation.

Attributes measured

- Oak species
- Number of seedlings/sprouts by size class
- Oak seedling/sprout type

Equipment and supplies

- Map showing savanna boundary, soil moisture type boundaries if applicable, and sampling point locations; satellite imagery or topography is helpful
- GPS unit preloaded with coordinates of sampling points or appropriate map if not using a GPS
- Survey protocol
- Datasheets or mobile device equipped for data entry
- Clipboard
- Pencils/pens
- List of species observed in prior surveys (if any), if desired
- Field references, such as field guides or cheat sheets (hard copy or digital)
- 4-foot fluorescent flags and/or fluorescent flagging tape
- Hand lens
- Compass
- Metric cloth measuring tape
- Surveyor's pins
- Plastic bags/cooler and/or plant press for specimen collection
- Camera to photograph unknown woody species

Survey timing and schedule

The oak seedling/sprout microplots should be implemented after any major restoration action such as tree thinning and thereafter at least every 3-5 years.

Sources of error

A brief list to consider is:

- Inconsistent application of survey methods
- Species misidentification
- Missing species

Establishment of sampling units

Each plot will be located using a GPS unit with preloaded coordinates. Use appropriate GPS protocols to navigate to the monumented center point of the tree macroplot. Oak seedling/sprout

microplots are circular, with a radius of 2.1m and a center offset 4 m to the east of the macroplot center monument. They are thus nested within the macro and tree sapling, shrub and vine microplots. Each macroplot has one oak seedling microplot. If using an alternative non-GPS method of locating points indicate this in the comments field, and make sure your methods are well documented.

Care should be used when monitoring near the boundary of adjacent savanna areas, because there may be a region of overlap (or a gap in application) of management in a unit. If savanna areas are highly interspersed (e.g., with bluff and lowland topography) or if management is applied in very patchy or complicated ways such that monitoring them separately would be impractical, they may be monitored together and treated as a single sample unit.

Data collection procedures

Using a handheld GPS, or map and compass, navigate to each monumented macroplot center. Visually define the circular sampling plot with flags or flagging tape. Conduct oak seedling microplot survey.

For all live oak seedlings and sprouts (< 2.5 cm DBH)) record:

- Species
- Number of stems by size class (< 0.9 m tall and > 0.9 m tall). Record the height class based on the tallest stem in the clump.
- Oak seedling/sprout type
 - new seedling - less than a foot in height; an acorn may be discovered by gently probing in the soil or by pulling one up that is not in the plot to use as a reference
 - less than a foot tall and less than a quarter inch or 10mm in basal stem diameter; leaves are often smaller than those on sprouts but that varies by the light environment with larger leaves developing in open full light conditions; see **Figure SOP 6.1**)
 - Sprout – young tree that has been clipped by a herbivore or mower etc., or its shoot has been killed by fire resulting in multiple branches or sprouts of new growth after the injury; the tree may be several years old and up to an inch in diameter, the new sprouts are often arising from a larger root system that has developed over the years. Height is less than 3 feet.



Figure SOP 6.1. One year-old pin oak seedling (left), note acorn still attached and intact original stem; bur oak sprout (center and right), note multiple older, dead stems in contrast to two live sprouts that emerged in the current year.

A simple user guide will be provided for electronic data entry using Survey123 or other application, and when available, it will be located in our [monitoring](#) ServCat site.

A sample blank datasheet and an example filled out datasheet are attached to this protocol ([Appendix](#)).

Data field details

OVERALL SITE: Record the name of the overall site.

MANAGEMENT UNIT: Record the name of the management unit (the larger savanna unit if there is one).

MONITORING AREA: Record the name of the savanna monitoring area. This can be a subunit of the larger management unit and is the area targeted by the oak seedling and sprout microplots.

SOIL MOISTURE TYPE: Record “dry,” “mesic,” or “wet,” as defined by the soil moisture categories within the monitoring area ([SOP 7](#)).

DATE: Record the date of the survey as yyyy/mm/dd.

OBSERVER(S): Record each observer’s name.

MACROPLOT NUMBER: Record the macroplot number.

MICROPLOT NUMBER: Record the microplot number.

SLOPE: Record the general slope of the microplot.

SOIL MOISTURE TYPE PER POINT: Document the soil moisture type (SOP 7) for each sampling point (plot number). Write a comment if what is indicated in the soil survey appears different from what is found on the ground.

OAK SPECIES: For all live oak seedlings <2.5 cm, DBH species using current scientific name in appropriate categories. Any abbreviations, codes or names used should be referenced to a master list of corresponding scientific names and the authority (source of the scientific names such as USDA Plants, Flora of North America, or ITIS) used. If you cannot identify a plant to species, record genus if possible and/or collect and/or photograph a specimen for future identification. For plants with enough information to distinguish from other species, keep track of unknowns by coding them discretely until they can be properly identified.

SEEDLING TYPE: Record oak seedling type (new seedling or sprout; (**Figure SOP 6.1**) for each oak seedling observed.

STEM COUNTS: Record counts of oak seedlings by species and by size class (< 0.9 m tall and > 0.9 m tall) for each species.

COMMENTS: Record any relevant comments, including presence of a mowed or disturbed area, citing the reason for disturbance if known, or concentrations of highly invasive species

A sample blank datasheet and **an example filled out datasheet** are attached to this protocol (Appendix C).

SOP 7: Using the NRCS Web Soil Survey to Identify Soil Drainage Classes Within a Survey Area

Note to testers: the following guidance was adapted from the National Protocol Framework for Monitoring in Prairie Reconstructions and will be included in the final guidance for savanna monitoring. It is included here so that you can understand how we will identify moisture categories. *If you have an alternative way to identify NRCS soil drainage classes to delineate areas tested, feel free to use that. Having the proper moisture category (wet, mesic, dry) is important to our analysis. Thank you!*

This SOP describes the basic steps for finding the soil drainage classes for a particular survey area and reducing them to the three soil moisture categories used in the PRI monitoring protocol.

- Section A lists steps to visualize the information using the Natural Resources Conservation Service (NRCS) soil survey website.
 - When you use the online tools, you can import a shapefile for one monitoring area, or you can digitize the outline of the monitoring area on a basemap.
 - If you wish to do all your GIS work in ArcMap (for example, you have a layer that outlines each of your monitoring areas), skip Section A, and download the soils map for your county as described in Section B.
 - If you already have a soils layer for your area with drainage class as one of the soil polygon properties, you can skip directly to Section C.
- Section B gives short guidance for downloading the map created in Section A for a particular area of interest (AOI; i.e., an individual monitoring area in this case). Alternatively, you can download the soils map for your county and do all the GIS work in ArcMap.
- Section C gives a brief description of steps that might be taken in ArcMap to process the information.

Keep in mind that the NRCS soil survey was mapped at a certain scale depending on when and where it was mapped. Therefore, boundaries between soil map polygons are approximations. When you examine the soil survey for a monitoring area, you may get a message that “you have zoomed in beyond the scale at which the soil map for this area is intended to be used.” We recognize that the extent and location of mapped soil units within each monitoring area are approximations. Nonetheless, the NRCS soil survey is a standard tool we can use to better understand how soil moisture type influences the outcomes of plantings.

A) Visualizing soil moisture categories online

- 1) Go to NRCS Soil Survey page:
<https://websoilsurvey.sc.egov.usda.gov/App/HomePage.htm>
- 2) Click the green circular “Start WSS” button, or click “Start Web Soil Survey” blue text in right sidebar.
- 3) Click the “Area of Interest (AOI)” tab to define the location for your map.
 - a) You can import a shapefile with one monitoring area or follow the steps 3b-3d to define a monitoring area using the NRCS web tools.
 - b) Zoom in on the map of the US, or use the State/County dropdowns on the left, and click “View” to zoom in; continue to zoom until your survey area more or less fills the view.
 - c) Use the polygon AOI tool or the rectangle AOI tool to outline your area of interest (i.e., monitoring area that you plan to monitor).
 - d) If you wish to remove streams, roads, etc. from the map, click the “Legend” tab at top left of the map, and uncheck the appropriate layer(s).
 - e) To define a new AOI for a subsequent map, click “Clear AOI,” then redo the above steps.
 - f) Note that you may download the soil survey for your county (Section B); then you would have the layer for your whole extent and could use a local copy of ArcMap to create individual maps for each monitoring area.
- 4) The following steps assume you have created an AOI for your monitoring area.
- 5) Click the “Soil Map” tab to see the outlines of the soil polygons within your AOI.
 - a) You can change the layout on your screen by choosing one of the two buttons just above the map and to the right—either normal map layout or full width map layout. With normal layout, the table with the soils descriptions is to the left of the map. With full width layout, it is below the map.
 - b) You can click on a soil name to see a description of that soil (map unit).
- 6) Click the “Soil Data Explorer” tab to visualize properties of the soils within your AOI, specifically drainage class.
 - a) Click on the “Soil Properties and Qualities” sub-tab.
 - b) Expand “Properties and Qualities Ratings” on the left sidebar (if not already expanded).
 - c) Further expand “Soil Qualities and Features.”
 - d) Click “Drainage Class.”
 - i) The Map and Table checkboxes should both be automatically checked. Dominant Condition should be automatically chosen as the aggregation method.
 - ii) Click the “View Rating” button to see your AOI with the soil polygons color coded by drainage class.
 - iii) Click the “Legend” tab at the upper left of the map to see which color represents which drainage class, and find the corresponding PRI moisture type (i.e., dry, mesic, or wet; **Figure SOP 7.1**). Note that the basemap can distort the colors in the polygons.
 - iv) Scroll down the page to see a table summary by map unit.

- (1) If your AOI outlines a monitoring area, you can use the percentages given in the table to determine how many plots need to be in each soil moisture category for that monitoring area.
- (2) To get this information into ArcMap, see steps B and C.
- v) **Figure SOP 7.2** depicts an example map generated from the NRCS Web Soil Survey site and composed of all mesic soils, so stratification is not needed for the monitoring protocol.
- vi) **Figure SOP 7.3** depicts a map with both wet and mesic soils. Monitoring for this monitoring area would be stratified by soil moisture type.
- 7) To download a shapefile of the soils map for your AOI, go to B).
- 8) Click “Shopping Cart (Free)” tab to download a report with a printed soils map of your area of interest, if desired.

PRI Soil Moisture Category	NRCS Soil Drainage Class
Dry	Excessively drained
	Somewhat excessively drained
Mesic	Well drained
	Moderately well drained
	Somewhat poorly drained
Wet	Poorly drained
	Very poorly drained
	Subaqueous
	Not rated or not available

Figure SOP 7.1. Correspondence of soil drainage classes from the NRCS Web Soil Survey (“Soil Rating Polygons” in legend with colored boxes) to soil moisture categories used in this protocol.

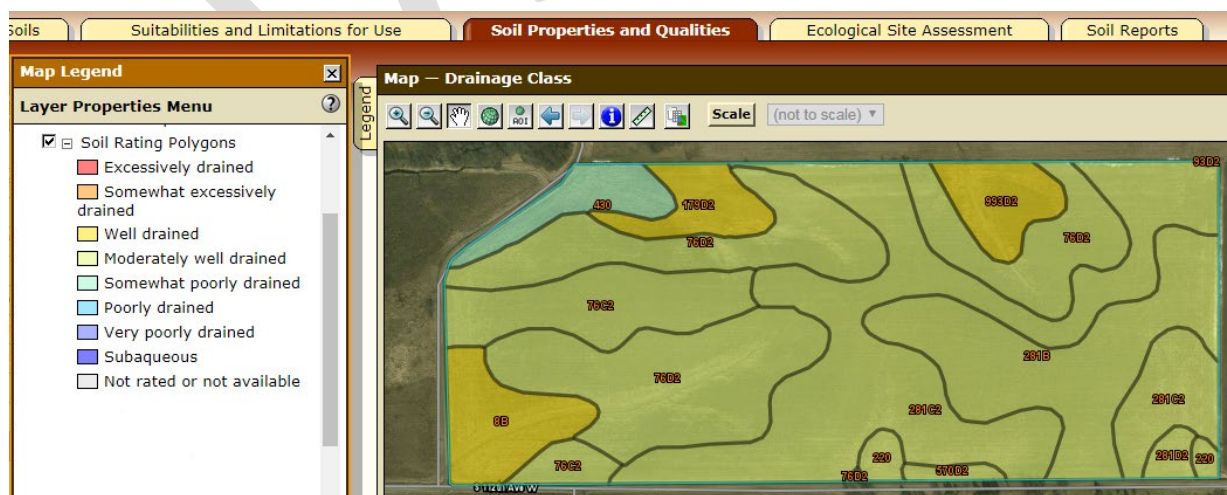


Figure SOP 7.2. An example of a soils map that was generated using the NRCS Web Soil Survey, with only one soil moisture category. The three soil drainage classes that occur in this monitoring area are all categorized as mesic, so this site would be monitored without stratification.

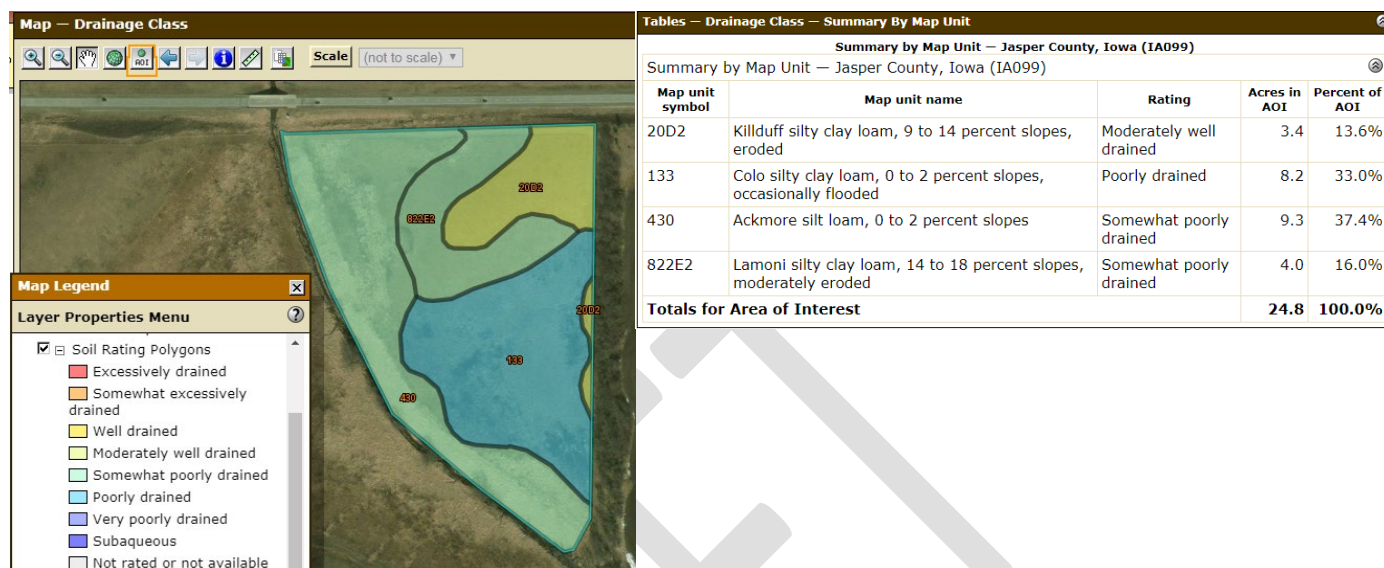


Figure SOP 7.3. An example of a soils map that was generated using the NRCS Web Soil Survey, with more than one soil moisture category. The poorly drained soil (which comprises 33% of the total area) is categorized as wet, while the other soils are mesic. For this monitoring area, monitoring would be stratified by soil moisture category. Species would be documented separately in the wet polygon for the meandering walk, and 33% of the nested frequency plots would be placed in the wet polygon while 67% of plots would be placed throughout the remainder of the planting. The area is 10 ha (24.8 acres) and would have 30 plots (3 plots/ha), 10 in wet and 20 in mesic.

B) Downloading soils data from NRCS Web Soil Survey

- 1) These steps assume that you followed Section A through Step 7 (and would like to download a soils map for your AOI) or that you want to download a county-wide soils map.
- 2) Click “Download Soils Data” tab to download a folder containing shapefiles and soils data.
- 3) You have three options for downloading data:
 - a) Your AOI (SSURGO)—this is a detailed soil survey for your previously defined area of interest.
 - b) Soil Survey Area (SSURGO)—this is a detailed soil survey that can be downloaded by county.
 - c) U.S. General Soil Map (STATSGO2)—this is a general soil survey that can be downloaded by state – however, it is not at a level of detail appropriate for a parcel-level view and should not be used for this protocol.
- 4) Click “Create Download Link” button to generate a link to a zip file with your data (link appears at the bottom of the page). When downloading a county map (3b), the link is automatically generated.

- 5) Once extracted from the zip file, the downloaded folder contains a number of items, including a “readme.txt” help file.
 - a) Shapefiles are in the “spatial” folder.
 - i) All spatial data are provided in Esri Shapefile format in WGS84 geographic coordinates. To be able to calculate area the data need to be transformed into a projected coordinate system such as UTM for the zone appropriate for your area (this can be done in ArcMap using “ArcToolbox--Data Management Tools--Projections and Transformations”).
 - ii) Codes used in layer names:
 - (1) soilmu = soil map unit
 - (2) soilsf = soil special feature
 - (3) _p_ = point
 - (4) _l_ = line
 - (5) _a_ = polygon
 - (6) soilsa_a_ = soil survey area boundary
 - (7) aoi_a_aoi = boundary of the area of interest
 - iii) The primary shapefile of interest for these purposes (the one that you need to work with in ArcMap) is “soilmu_a_****”, the polygon layer with the soil map units.
 - iv) A soil map unit is a collection of areas (i.e., polygons) defined and named the same in terms of their soil components (e.g., Tama silty clay loam 2 to 5% slopes). See NRCS documentation for more detail.
 - b) Data tables are in the “tabular” folder.
 - i) Double click on the Access database template in the main folder, and follow the instructions to import the data tables into a usable format.
 - ii) The “muaggatt” table contains information about soil drainage class.
 - c) Consult the “readme.txt” file for additional guidance in downloading and using the NRCS soils data.

C) Using ArcMap to incorporate soils data

- 1) Make sure the soils layer and any other layers you need are in the same projection. Add a basemap if desired.
 - a) As indicated above (B.4.a.i.), if a spatial data layer is in geographic coordinates, it needs to be transformed into projected coordinates.
- 2) Add a column to the attribute table that contains the area for each soil polygon.
 - a) Open the attribute table for your projected layer. The field Shape_Area is in square meters. Using “Table Options--Add field,” create a new field named Hectares (or Acres, if preferred; however hectares are used to calculate the number of plots needed), and choose type “float” (has decimal places, takes up less room than “double”).
 - b) A new empty column will be appended.
 - c) Right click on the heading and choose “Calculate Geometry.”
 - i) Choose the desired units to be calculated. Uncheck the box “Calculate selected records only.” Click “OK.”

- ii) It seems that the calculation cannot be done on a joined table, so do this prior to joining to the soil moisture table, or unjoin the tables, perform the calculation, and then rejoin.
- 3) You can add the “muaggatt” table in the downloaded Access database to ArcMap; however, the database is read only, making it difficult to create and edit the needed new field that assigns drainage class to the appropriate soil moisture category (dry, mesic, wet).
- 4) This is one workaround for getting only the necessary information into ArcMap (there may be other, more efficient ways to achieve this).
 - a) We are interested in table “muaggatt” and its field “drclassdcd,” or drainage class-dominant condition, which gives the most common drainage class by area for each soil map unit. The procedure below will work for either of these cases:
 - i) If you downloaded data for an entire county, the tables contain records for all soils found in the county.
 - ii) If you downloaded data for an area of interest, the tables contain records only for soils found in that delineated area. (However, if you generate a report, the acreages are for the entire county.)
 - b) Open the “muaggatt.txt” file in Excel as a delimited file. We only really need the “drclassdcd” column (16th column, P) and the “mukey” column (last column, AN); so delete the other columns. However, you may find it informative to keep the map unit name (“muname,” 2nd column) or other information.
 - i) To view the other column names, open the table in Access (step B.3.b.).
 - ii) Insert a row for the column headers, and type them in. Use names with 10 or fewer characters.
 - c) Using Excel, assign values to a new column (give it an appropriate heading such as SoilMoist) using a nested IF command that references the appropriate cell.
Example formula:
 - i) =IF(B2="Excessively drained","Dry",IF(B2="Somewhat excessively drained","Dry",IF(B2="Well drained","Mesic",IF(B2="Moderately well drained","Mesic",IF(B2="Somewhat poorly drained","Mesic",IF(B2="Poorly drained","Wet",IF(B2="Very poorly drained","Wet","Null"))))))))
 - ii) In this case, the formula was going into line 2 of the SoilMoist column, and the drainage class was in column B; the formula was then copied and pasted to the other cells in the column.
 - iii) In general, if drclassdcd is not one of the listed values, it is a permanently-flooded or other non-soil feature.
 - d) The “mukey” column in the Excel file needs to be formatted as text in order to match the field type of “MUKEY” in the soils layer.
 - e) Save as an Excel workbook, and import the table into ArcMap. Join to the soil map layer on MUKEY/mukey.
 - i) When you join, it is apparently more efficient for future performance to “Keep only matching records.”
 - ii) Prior to joining the layers, perform an area calculation as described in step 2.

- 5) In “Properties” you can create labels for the polygons and/or change the symbology (i.e., color differently) according to soil moisture category to visualize the three categories within your area of interest.
 - a) You can also use labels or color to visualize the distribution of the original seven NRCS drainage classes.
- 6) If desired, you can dissolve the soil polygon boundaries of adjacent like polygons. However, maintaining those boundaries can be informative when using the map in the field.
- 7) Optionally, to include additional soils information, follow these steps.
 - a) Open the Access database template (B.3.b.), and add it to the map. You only need the “muaggatt” table to create maps based on soil drainage class, but you may include other tables if you are interested in that information. See the readme.txt help files in the downloaded zip file for more guidance.
 - b) Right click on the “soilmu_a_*****” layer, and create a join to any of the tables in the database using the variable “MUKEY.”
- 8) Compute the total area within each moisture category in your AOI. Summarize each monitoring area separately.
 - a) Open the attribute table of the projected layer for your AOI, which now has hectares (or acres) and soil moisture category for each soil polygon.
 - i) If the summary below fails (the join seems to interfere with the summarization), export the attribute table to a text file, add it to the map, and proceed with these steps.
 - b) Right click on the soil moisture heading, and choose “Summarize.”
 - c) Expand Hectares in 2, and choose “Sum” to get total hectares for each soil moisture category.
 - d) Save, and add to the map. If the save fails, you may have to change the file type.
- 9) For a given monitoring area, find the number of plots needed in each soil moisture category.
 - a) To get the proportion of each moisture category, take the hectares for that category, and divide by the total hectares of the monitoring area.
 - b) To determine how many plots should be in each category, take the above proportion, and multiply by the total number of plots (3 plots/ha but no fewer than 10 and no more than 40; see Elements 2 and 3). Round to the nearest integer.

Note: Instructions for the NRCS soil survey website are accurate as of 7/12/2019.

SM 1: Monitoring Macroplot Site Diagram

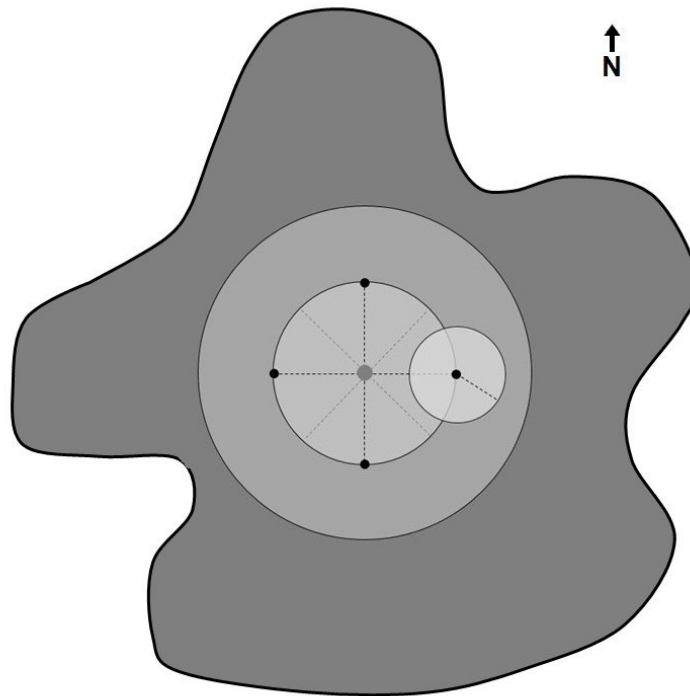


Diagram Key (Listed by size)

- | | |
|---|---|
|  | SOP 3-Tree Macroplot <ul style="list-style-type: none">• Size determined by prism |
|  | SOP 5-Tree sapling, Shrub, and Vine Microplot <ul style="list-style-type: none">• 7.3m radius around Macroplot center monument |
|  | SOP 4-Nested Frequency Plots <ul style="list-style-type: none">• 4m N, E, S, and W of the Macroplot center monument.• Optional: add 4 additional nested frequency plots NW, NE, SW, and SE of the Macroplot center monument. |
|  | SOP 6-Oak Seedling Microplot <ul style="list-style-type: none">• 2.1m radius around the E nested frequency plot.• Optional: add 3 additional Oak Seedling Microplots, one at each of the remaining Nested Frequency Plots |
- Tree macroplot center monument (SOP 3)
- Nested frequency plot sampling unit (SOP 4)

SM 2: Monitoring Protocol Checklist

1. Mark the Macroplot (SOP 3) center monument with a survey pin
 - a) Assign a unique Macroplot ID to the site. You will refer to this ID on each of the SOP datasheets at this Macroplot
 1. For example, if your savanna unit is called “Thorn Valley”, you could make your Macroplot ID “TV-1”. The next Macroplot could be named “TV-2” (if you’re monitoring within the same savanna unit).
 - b) Document the Macroplot center monument GPS coordinates on the data sheet
 - c) Secure the end of the measuring tape with the pin
2. Using the measuring tape, set up markers (flags) at 4m for **SOP4** and 7.3m for **SOP5** in each cardinal direction from center marker (as illustrated in Appendix A: Savanna Monitoring Macroplot Site Diagram)
3. Stand at the center of the Macroplot and take photos in each of the 4 cardinal directions
 - a) Repeating the photo order at every Macroplot center monument is suggested
4. Begin data collection with **SOP4: Nested Frequency Plots** – start at the north 4m marker (flag)
5. Continue with **SOP4: Nested Frequency Plots** at the east 4m marker
6. Conduct **SOP6: Oak Seedling Microplot** survey
7. Continue **SOP4: Nested Frequency Plots** at South and West 4m markers
8. Conduct **SOP5: Tree Sapling, Shrub and Vine Microplot** within the 7.3m boundaries of the plot (divide the microplot area between observers, if possible, to prevent duplication of data collected)
9. Conduct **SOP3: Tree Macroplots** using 2m basal area factor (BAF) prism to determine trees to be sampled. Record data:
 - a) Species
 - b) DBH
 - c) Tree Form
 - d) Crown Condition
 - e) Tree Live Crown Ratio
 - f) Crown Light Exposure
 - g) Basal Fire Scars
10. Complete SOP3 by recording Macroplot data:
 - a) Soil Moisture Type
 - b) Aspect
 - c) Slope
 - d) Canopy Coverage Percent (N, E, S, W)
11. Verify that the base data (date, plot ID, observer, etc.) **has been entered on each survey datasheet**
12. Conduct **SOP1 and 2: Meandering Walk** annually or near-annually in spring and/or summer seasons

SM 3: Monitoring Datasheets

DRAFT

Datasheet for Meandering Walk (SOP1)-Stratified by Soil Moisture Type

Overall site: _____ **Management unit:** _____ **Monitoring area:** _____

Soil moisture type: _____ Date: _____ Observer(s): _____ Datum _____

Start time: _____ End time: _____ Total time: _____ Area searched (hectares): _____

Record species encountered, a general abundance category for each species in each soil moisture type based on the number of individuals encountered on the walk, and whether the individuals were localized or widespread.

[illegible]**Comments:**

Datasheet for *Stratified* Meandering Walk (SOP1) - Continued

Date: _____ **Monitoring area:** _____

[illegible]**Comments:**

Datasheet Example for Meandering Walk (SOP1)-Stratified by Soil Moisture Type

Overall site: Neal Smith NWR Management unit: Thorn Valley Monitoring area: TV - 1

Soil moisture type: Mesic Date: 6/10/21 Observer(s): K. Viste-Sparkman & K. Jacobs Datum _____

Start time: 09:15 End time: 11:35 Total time: 2 hrs 20 mins Area searched (hectares): 3

Record species encountered, a general abundance category for each species in each soil moisture type based on the number of individuals encountered on the walk, and whether the individuals were localized or widespread.

Species Name	DRY		MESIC		WET	
	Abundance (Sparse, Occasional, Frequent, Very Common)	Distribution (Localized, Widespread)	Abundance (Sparse, Occasional, Frequent, Very Common)	Distribution (Localized, Widespread)	Abundance (Sparse, Occasional, Frequent, Very Common)	Distribution (Localized, Widespread)
<i>Bromus inermis</i>	-	-	F	L	-	-
<i>Quercus macrocarpa</i>	-	-	F	W	-	-
<i>Monarda fistulosa</i>	-	-	S	L	-	-
<i>Solidago Canadensis</i>	-	-	VC	W	-	-
<i>Rhus glabra</i>	-	-	F	W	-	-
<i>Helianthus tuberosa</i>	-	-	O	L	-	-
<i>Vitis riparia</i>	-	-	F	W	-	-
<i>Cirsium discolor</i>	-	-	S	W	-	-
<i>Ulmus rubra</i>	-	-	S	L	-	-
<i>Rubus occidentalis</i>	-	-	F	W	-	-
<i>Symphoricarpos orbiculatus</i>	-	-	VC	W	-	-
<i>Zanthoxylum americanum</i>	-	-	O	L	-	-
<i>Viola sororia</i>	-	-	O	W	-	-
<i>Celtis occidentalis</i>	-	-	S	L	-	-
<i>Impatiens capensis</i>	-	-	S	L	-	-
<i>Eutrochium purpureum</i>	-	-	F	L	-	-
<i>Aesculus glabra</i>	-	-	F	L	-	-
<i>Polygonatum biflorum</i>	-	-	S	W	-	-
<i>Phlox divaricata</i>	-	-	S	W	-	-
<i>Ratibida Pinnata</i>	-	-	S	L	-	-

Comments:

Datasheet for Woody Species Meandering Walk (SOP 2) *one soil moisture type*

Date: _____ Observer(s): _____ Overall site: _____ Management unit: _____

Monitoring name: _____ Polygon name: _____ Soil moisture type: _____

Start time: _____ End time: _____ Total time: _____ Area searched (hectares): _____ Datum _____

For each sample point stop, record sample point number, GPS location, tree species encountered, basal area, average tree diameter and maximum/minimum oak tree diameter observed by species. Record herbaceous percent cover and cover categories, tree canopy cover, and shrub/vine cover category. Tally oak trees in each condition category. If multiple days are used to record data, use separate data sheets to include the date.

Sample Point Number	Sample Point (GPS)	Herbaceous abundance	Tree Abundance	Tree Type	Shrub Abundance	Tree Canopy Cover %	Oak Condition	Oak Seedling/Sprout Abundance

Comments:

Woody Species Meandering Walk *continued*

Date: _____ Monitoring name: _____ Polygon name: _____ p. _____ of _____

Sample Point Number	Sample Point (GPS)	Herbaceous abundance	Tree Abundance	Tree Type	Shrub Abundance	Tree Canopy Cover %	Oak Condition	Oak Seedling/Sprout Abundance

Comments:

Datasheet Example for Woody Species Meandering Walk (SOP 2) *one soil moisture type*

Date: 10/15/20 Observer(s): K. Jacobs & T. Turner Overall site: Neal Smith NWR Management unit: Thorn Valley
 Monitoring name: _____ Polygon name: _____ Datum _____ Soil moisture type: Mesic Start time: 09:44
 End time: 13:14 Total time: 3 hours 30 min Area searched (hectares): 4

For each sample point stop, record sample point number, GPS location, tree species encountered, basal area, average tree diameter and maximum/minimum oak tree diameter observed by species. Record herbaceous percent cover and cover categories, tree canopy cover, and shrub/vine cover category. Tally oak trees in each condition category. If multiple days are used to record data, use separate data sheets to include the date.

Sample Point Number	Sample Point (GPS)	Herbaceous abundance	Tree Abundance	Tree Type	Shrub Abundance	Tree Canopy Cover %	Oak Condition	Oak Seedling/Sprout Abundance
1	41.555714° N 93.268456° W	Moderate	9		Dense	60%	Intermediate	1
2	41.555511° N 93.265265° W	Moderate	10		Moderate	65%	Intermediate	-
3	41.555364° N 93.265045° W	Dense	4		Moderate	30%	Healthy	2
4	41.555216° N 93.265772° W	Dense	5		Moderate	45%	Intermediate	1
5	41.554665° N 93.265588° W	Moderate	11		Sparse	75%	Intermediate	-
6	41.554793° N 93.266237° W	Sparse	14		Moderate	84%	Unhealthy	-

Oak Species Health and Form Datasheet for Woody Species Meandering Walk - (SOP 2)

Date: _____ Observer(s): _____ Overall site: _____
Management unit: _____ Monitoring name: _____ Polygon name: _____ Datum _____
Soil moisture type: _____ Start time: _____ End time: _____
Total time: _____ Area searched (hectares): _____

For each sample point stop, record sample point number, GPS location, and oak species encountered. Tally oak trees in each condition category. If multiple days are used to record data, use separate data sheets to include the date.

		Tree Health Condition				Branching form	
Sample Point Number:	Oak Species	Healthy	Intermediate	Unhealthy	Dead	Upward Branching	Outward Branching
GPS:							

Comments:

Oak Species Health and Form Datasheet for Woody Species Meandering Walk - (SOP 2)

Date: _____ Monitoring name: _____ Polygon name: _____ p. ____ of ____

For each sample point stop, record sample point number, GPS location, and oak species encountered. Tally oak trees in each condition category. If multiple days are used to record data, use separate data sheets to include the date.

		Tree Health Condition				Branching form	
Sample Point Number:	Oak Species	Healthy	Intermediate	Unhealthy	Dead	Upward Branching	Outward Branching
GPS:							

Comments:

Oak Species Health and Form Datasheet for Woody Species Meandering Walk - (SOP 2)

Date: 10/12/20 Observer(s): T. Turner & K. Viste-Sparkman Overall site: Neal Smith NWR

Management unit: Thorn Valley Monitoring name: _____ Polygon name: _____ Datum: _____

Soil moisture type: Mesic Start time: 9:26 End time: 11:56 Total time: 2hrs 20min Area searched (hectares): 2

For each sample point stop, record sample point number, GPS location, and oak species encountered. Tally oak trees in each condition category. If multiple days are used to record data, use separate data sheets to include the date.

		Tree Health Condition				Branching form	
Sample Point Number:	Oak Species	Healthy	Intermediate	Unhealthy	Dead	Upward Branching	Outward Branching
GPS: 41.555714°N 93.268456°W	<i>Q. macrocarpa</i>	X					X
	<i>Q. rubra</i>		X			X	
41.555511°N 93.265265°W	<i>Q. alba</i>				X	-	-
	<i>Q. alba</i>		X			X	
41.555364°N 93.265045°W	<i>Q. rubra</i>	X					X
	<i>Q. alba</i>	X					X
41.554793°N 93.266237°W	<i>Q. macrocarpa</i>			X		X	
	<i>Q. macrocarpa</i>		X				X

Comments:

Overstory Tree Macroplot Datasheet (SOP 3)(trees ≥ 12.7 cm DBH)

Date: _____ Observer(s): _____ Overall Site: _____ Management Unit: _____ Monitoring Area: _____

Macroplot ID: _____ Macroplot GPS: _____ Soil Moisture Type: _____ Aspect: _____ Slope: _____ Datum: _____

Photo Point: _____ Canopy Cover %: N: _____ E: _____ S: _____ W: _____

Tree number	Species	DBH ¹	Crown Condition ²	Tree Form ³	Tree Height to Live Crown Ratio ⁴	Tree Crown Light ⁵	Fire Scar					
							Height ⁶	Width	Depth	Tree Circumference at Scar ⁷	Scar Orientation ⁸	Scar Type ⁹

¹Diameter at breast height (1.37m) with diameter tape (to the nearest 0.5 cm) of tree

²Codes: 1=Healthy, 2=Light decline, 3=Moderate decline, 4=Severe decline, 5=Dead

³Codes: O=Outward, U=Upward growth, S=Shade pruned

⁴Ratio of the height of live crown to the total tree height. Categories: <10 = 1, 10-25 = 2, 26-50 = 3, 51-75 = 4, >75 = 5

⁵The amount of direct sunlight a tree receives when the sun is directly overhead. Codes: 0=no full light because it is completely shaded, 1=tree receives full light from the top *or* 1 side, 2=top and 1 side, 3=top and 2 sides, 4=top and 3 sides, 5=top and 4 sides

⁶Measurements taken to the nearest 0.5 cm.

⁷Tree circumference is taken at widest part of scar.

⁸Orientation is NE, N, NW, W, SW, S, SE, E.

⁹Scar Type is F=fire; M=mechanical.

Comments:

Datasheet for Overstory Tree Macroplot Datasheet (SOP 3) – *Continued* (trees ≥ 12.7 cm DBH)

Macroplot ID: _____ Date: _____

Tree number	Species	DBH ¹	Crown Condition ²	Tree Form ³	Tree Height to Live Crown Ratio ⁴	Tree Crown Light ⁵	Fire Scar					
							Height	Width	Depth	Tree Circumference at Scar ⁷	Scar Orientation ⁸	Scar Type ⁹

¹Diameter at breast height (1.37m) with diameter tape (to the nearest 0.5 cm) of tree

²Codes: **1**=Healthy, **2**=Light decline, **3**=Moderate decline, **4**=Severe decline, **5**=Dead

³Codes: O=Outward, U=Upward growth, S=Shade pruned

⁴Ratio of the height of live crown to the total tree height. Categories: $<10 = 1$, $10-25 = 2$, $26-50 = 3$, $51-75 = 4$, $>75 = 5$

⁵The amount of direct sunlight a tree receives when the sun is directly overhead. Codes: **0**=no full light because it is completely shaded, **1**=tree receives full light from the top or 1 side, **2**=top and 1 side, **3**=top and 2 sides, **4**=top and 3 sides, **5**=top and 4 sides

⁶Measurements taken to the nearest 0.5 cm.

⁷Tree circumference is taken at widest part of scar.

⁸Orientation is NE, N, NW, W, SW, S, SE, E.

⁹Scar Type is F=fire; M=mechanical.

Comments:

Datasheet Example for Overstory Tree Macroplot Datasheet (SOP 3) (trees ≥ 12.7 cm DBH)

Overall Site: Neal Smith NWR. Management Unit: Thorn Valley Monitoring Area: TV – 1. Macroplot ID: 1.

Macroplot GPS: 41.335231°N 93.161012°W Soil Moisture Type: Mesic Aspect: 269 Slope: 44% Datum: _____

Date: 10/06/20 Observer(s): K. Viste-Sparkman, K. Jacobs, T. Turner Photo Point: _____ Canopy Cover %: 65% N: 60 E: 25 S: 83 W: 80

Tree number	Species	DBH ¹	Crown Condition ²	Tree Form ³	Tree Height to Live Crown Ratio ⁴	Tree Crown Light ⁵	Fire Scar					
							Height ⁶	Width	Depth	Tree Circumference at Scar ⁷	Scar Orientation ⁸	Scar Type ⁹
1	<i>Q. macrocarpa</i>	70 cm	1	S	> 75	4	70.5cm	20.4 cm	1cm	250cm	SW	F
2	<i>Q. macrocarpa</i>	109 cm	2	S	>75	4	38cm, 41 m, 10.8cm	10cm, 35cm, .8cm	1.4cm, 7.5cm, 3cm	402cm, 402cm, 387cm	NW, S, NW	M, M, F
3	<i>Q. macrocarpa</i>	94 cm	3	S	>75	4	59.5cm, 10cm, 18cm	33.4 cm, 5.5cm, 22cm	2.5cm, 6.2cm, 1.5cm	264.5cm, 301cm, 264.5cm	SE, N, W	F, M, F
4	<i>Q. macrocarpa</i>	76.4 cm	4	S	51-75	4						
5	<i>Q. macrocarpa</i>	109.5 cm	1	S	51-75	5						
6	<i>Q. macrocarpa</i>	81.5 cm	5	S	<10	4						

¹Diameter at breast height (1.37m) with diameter tape (to the nearest 0.5 cm) of tree

²Codes: 1=Healthy, 2=Light decline, 3=Moderate decline, 4=Severe decline, 5=Dead

³Codes: O=Outward, U=Upward growth, S=Shade pruned

⁴Ratio of the height of live crown to the total tree height. Categories: <10 = 1, 10-25 = 2, 26-50 = 3, 51-75 = 4, >75 = 5

⁵The amount of direct sunlight a tree receives when the sun is directly overhead. Codes: 0=no full light because it is completely shaded, 1=tree receives full light from the top or 1 side, 2=top and 1 side, 3=top and 2 sides, 4=top and 3 sides, 5=top and 4 sides

⁶Measurements taken to the nearest 0.5 cm.

⁷Tree circumference is taken at widest part of scar.

⁸Orientation is NE, N, NW, W, SW, S, SE, E.

⁹Scar Type is F=fire; M=mechanical.

Comments: Tree 2 has a branch cavity, a trunk cavity, and a basal cavity (which we assume continues very high up the tree. The Trunk cavity diameter is 1cm. The questionable basal cavity was called the "largest."

Fill in the soil moisture type for each plot (SOP 3). Record the smallest subplot (A–E) within which each species occurred in each plot (**Figure 3.1**). Do not forget to “flip” the plot frame for the final subplot.

Nested plot Dimensions: (A) _____ (B) _____ (C) _____ (D) _____ (E) _____

[illegible]

Datasheet Example for Nested Frequency Plots (SOP 4)

Fill in the soil moisture type for each plot (SOP 3). Record the smallest subplot (A–E) within which each species occurred in each plot (**Figure 3.1**). Do not forget to “flip” the plot frame for the final subplot.

Date: 06/20/2018 Observers: A McColpin Overall Site: MySpot Monitoring Area: Unit A Macroplot ID: A-1

GPS Macroplot: _____ GPS Microplot: _____ Datum: _____

Soil Moisture type(s): Dry-63%/Mexic-37%

Nested plot dimensions (cm): (A) 12.5 x 50 (B) 25 x 50 (C) 50 x 50 (D) 50 x 100 (E) 50 x 200

	Plot Number (minimum 4, maximum 8)								Comments
	1	2	3	4	5	6	7	8	
Cardinal Direction from Center Monument:	N	E	S	W					
☒ Species									
(Planted Species)									
<i>Andropog. gerar.</i>	A	--	D	D					
<i>Antennaria negl.</i>	C	E	--	--					
<i>Boutel. curtipen.</i>	A	B	--	A					
<i>Coreops. palmata</i>	D	--	B	E					
<i>Dalea candida</i>	--	B	--	E					
<i>Dalea purpurea</i>	--	--	--	--					
<i>Echinac. pallida</i>	B	--	B	B					
<i>Elymus canade.</i>	--	A	--	E					
<i>Eryngium yucc.</i>	--	E	A	--					
<i>Helianthus occid.</i>	--	--	--	--					
<i>Heliopsis helian.</i>	A	E	C	E					
<i>Lespedez. capit.</i>	--	D	E	C					
<i>Liatris aspera</i>	--	C	D	--					
<i>Ratibida pinna.</i>	--	B	A	--					
<i>Rudbeckia hirta</i>	A	A	--	D					
<i>Schiza. scopari.</i>	D	C	E	A					
<i>Solidago nemor.</i>	--	E	--	C					
<i>Sporobolus heter.</i>	E	B	D	--					
(Additional Spp.)									
<i>Setaria viridis</i>	A	A	B	E					
<i>Trifolium repens</i>	B	--	--	--					
<i>Brassica nigra</i>	E	B	--	--					
<i>Taraxacum off.</i>	--	C	A	--					
<i>Oxalis stricta</i>	--	--	--	B					

(trees ≥ 2.5 cm DBH but < 12.7 cm DBH)

Date: _____ Observer(s): _____ Overall Site: _____
 Management Unit: _____ Monitoring Area: _____ Macroplot ID: _____ Datum: _____
 GPS Macroplot: _____ GPS Microplot: _____ Soil Moisture Type: _____

[illegible]

¹Saplings include any species between 2.5 and 12.7 cm DBH

²Shrubs are any species with a DBH less than 2.5 cm DBH. Measurements are taken to the nearest 0.5 m

Comments:

Datasheet for Tree Sapling, Shrub and Vine Microplot (SOP 5) *continued*

Date: _____ Monitoring Area: _____ Macroplot ID: _____ p. __ of __

Microplot Number	Species	Sapling Count ¹ by DBH Class:		Average Shrub Height ²	Shrub & Vine Percent Cover ³
		2.5 – 7.5 cm	7.6 – 12.7 cm		

¹Saplings include any species between 2.5 and 12.7 cm DBH
²Shrubs are any species with a DBH less than 2.5 cm DBH. Measurements are taken to the nearest 0.5 m

Comments:

Datasheet Example for Tree Sapling, Shrub and Vine Microplot (SOP 5)

(trees ≥ 2.5 cm DBH but < 12.7 cm DBH)

Date: 10/7/2020 Observer(s): K. Jacobs, T. Turner, K. Viste-Sparkman Overall Site: Neal Smith NWR

Management Unit: Thorn Valley Monitoring Area: Macroplot ID: TV - 2 Datum: Soil Moisture Type: Mesic

GPS Macroplot: 41.335231°N 93.161012°W GPS Microplot: 41.555511°N 93.265265°W

Microplot Number	Species	Sapling Count ¹ by DBH Class:		Average Shrub Height ²	Shrub & Vine Percent Cover ³
		2.5 – 7.5 cm	7.6 – 12.7 cm		
	<i>Quercus macrocarpa</i>			.5	1
	<i>Juglans nigra</i>			-	-
	<i>Prunus serotina</i>			1	1
	<i>Carya cordiformis</i>			-	3
	<i>Symphoricarpos orbiculatus</i>			1	1
	<i>Ribes missouriense</i>			1	1
	<i>Smilax tamnoides</i>			1	2
	<i>Carya ovata</i>			2.5	1
	<i>Rosa multiflora</i>			1.5	1
	<i>Rubus occidentalis</i>			1	1
	<i>Vitis riparia</i>			2	1
	<i>Celastrus scandens</i>			1.5	1
	<i>Rhus glabra</i>			1.75	1
	<i>Cornus racemosa</i>			1.25	
	<i>Lonicera sp.</i>			2.25	
	<i>Toxicodendron radicans</i>			.25	

¹Saplings include any species between 2.5 and 12.7 cm DBH

²Shrubs are any species with a DBH less than 2.5 cm DBH. Measurements are taken to the nearest 0.5 m

Comments:

Date: _____ Observer(s): _____ Overall Site: _____

Management Unit: _____ Monitoring Area: _____ Soil Moisture Type: _____ Datum: _____

Macroplot ID: _____ GPS Macroplot: _____ GPS Microplot: _____

[illegible]**Comments:**

Oak Tree Seedling & Sprout Microplot (SOP 6) *continued* (trees < 2.5 cm DBH)

Date: _____ Monitoring Area: _____ Macroplot ID: _____ p. ____ of ____

Microplot ID ¹	Soil Moisture Type at Point	Oak Species	Oak Seedling Type	Count by Height Class:	
				< 0.9 m tall	≥ 0.9 m tall

¹The Macro & Microplot ID is a combination of the Macroplot ID and the cardinal direction from the Macroplot center monument at which you are collecting data (e.g. TV-4 East or TV-4 E)

Comments:

Comments:

Appendix D – Supplemental Materials

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SOP 1-2 Field Reference: Meandering Walk Definitions

Abundance Category	
Sparse	Seldom occurring and usually occurring as an isolated plant; few total individuals observed
Occasional	Scattered plants in low numbers; sometimes occur in small clumps
Frequent	Many plants observed; species is regularly seen but rarely becomes dominant
Very Common	Species occurring in large numbers; can be dominant
Distribution Category	
Localized	Found in one or a few discrete spots in the area searched
Widespread	Dispersed broadly throughout the area searched
Herbaceous Cover Category	
Bare	Mostly bare soil without herbaceous vegetation present though evidence of bits of vegetation widely spaced (>1.5 m) may be present.
Sparse	Vegetation is depauperate, occurring as small, widely scattered clumps (0.9 to 1.5 m apart or more) or foliage is scant and growing in poor condition in low density; not contiguous
Moderate	Vegetation is regularly seen and may be somewhat contiguous but has notable breaks
Dense	Vegetation is fairly contiguous and generally without breaks and tends to be robust
Shrub/Vine Cover Category	
Sparse	Isolated individual shrubs; less than 5% cover of the area
Moderate	Clumps of shrubs or vines; approximately a third of the area occupied
Dense	Shrubs and vines abundant, approximately half or more of the area occupied
Oak Condition Category	
Outward branching	Some limbs grow in a predominantly horizontal pattern or reaching downward; overall growth pattern may appear somewhat rounded or formerly horizontal branches have been shade pruned as evidenced by limb scars and stumps (Figure SOP 1.2)
Upward branching	Branches mostly reach upward in a more vertical growth pattern reaching for sunlight; mostly occurs in areas that have been growing in densely wooded areas (Figure SOP 1.3)
Healthy	Leaves are green and abundant; trunk is intact
Unhealthy	Evidence of leaf discoloration, malformation or loss is evident; perforations or sloughing of bark, burn scars, notable portions of the tree are dead; concentrations of small twigs occur along major limbs

DRAFT